

Supplementary Information for

Multicomponent Thiolactone-Based Ionizable Lipid Screening Platform for Efficient and Tunable mRNA Delivery to the Lungs

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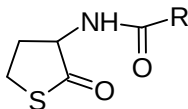
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Synthesis of ionizable lipids and their intermediates

General synthesis of intermediate (Bj).



DL-homocysteine thiolactone hydrochloride (1.8 mmol) was dissolved in 5 mL of anhydrous dichloromethane at room temperature. Then triethylamine (1.8 mmol) was added followed by EDC hydrochloride (1.3 mmol), 4-(dimethylamino)pyridine (0.20 mmol) and carboxylic acid (1.0 mmol). The reaction mixture was stirred at room temperature overnight under an argon atmosphere. The organic layer was dried with anhydrous MgSO_4 , filtered, and evaporated under reduced pressure. The resulting residue was purified by flash chromatography (gradient of hexane/ethyl acetate: 100/0 to 0/100) to afford compound **B** as a pure product.

The pure compounds were characterized by mass spectroscopy.

B1: Theoretical $[\text{M}+\text{H}]^+ = 384.29$, experimental $[\text{M}+\text{H}]^+ = 384.47$

B2: Theoretical $[\text{M}+\text{H}]^+ = 356.26$, experimental $[\text{M}+\text{H}]^+ = 356.44$

B3: Theoretical $[\text{M}+\text{H}]^+ = 382.28$, experimental $[\text{M}+\text{H}]^+ = 382.45$

B4: Theoretical $[\text{M}+\text{H}]^+ = 300.19$, experimental $[\text{M}+\text{H}]^+ = 300.32$

B5: The synthesis of this compound requires one previous oxidation step to obtain the respective carboxylic acid. First, 2-octyl-1-dodecanol (1077 mg, 3.5 mmol) and periodic acid (4144 mg, 18 mmol) were dissolved in acetonitrile (28 ml). Then pyridinium chlorochromate (23 mg, 0.030 mmol) was added to the previous solution. The reaction mixture was stirred 16 hours at room temperature. Then, the solvent was evaporated under reduced pressure. Thus, the crude was dissolved in AcOEt and first washed twice with water and finally with brine. The organic phase was dried with anhydrous MgSO_4 , filtered and evaporated under reduced pressure. The residue was used in the next step without any further purification.

Theoretical $[\text{M}+\text{H}]^+ = 412.32$, experimental $[\text{M}+\text{H}]^+ = 412.50$

General synthesis of non-commercial acrylates (Ck).

For those acrylates not available commercially the following general synthesis was followed:

Acryloyl chloride (4.0 mmol) was added dropwise to a cooled (0 °C) solution of alcohol (4.4 mmol) and triethylamine (6.0 mmol) in anhydrous dichloromethane (15 mL). The resulting solution was protected from light and stirred overnight under an argon atmosphere, allowing it to warm to room temperature. The reaction mixture was followed to completion by TLC chromatography. Then, the reaction mixture was first washed twice with water and finally with brine. The organic phase was dried with anhydrous MgSO_4 , filtered and evaporated under reduced pressure. The resulting residue was purified by

flash chromatography (gradient of hexane/ethyl acetate: 100/0 to 0/100) to afford the branched acrylate as a pure product.

C1: ^1H NMR (400 MHz, CDCl_3) δ : 6.39 (dd, $J = 17.4, 1.5$ Hz, 1H), 6.12 (dd, $J = 17.3, 10.5$ Hz, 1H), 5.81 (dd, $J = 10.4, 1.5$ Hz, 1H), 5.40 – 5.29 (m, 2H), 4.14 (t, $J = 6.7$ Hz, 2H), 2.01 (q, $J = 6.5$ Hz, 4H), 1.74 – 1.60 (m, 2H), 1.40 – 1.24 (m, 22H), 0.92 – 0.84 (m, 3H).

C3: ^1H NMR (400 MHz, CDCl_3) δ : 6.39 (dd, $J = 17.4, 1.5$ Hz, 1H), 6.12 (dd, $J = 17.3, 10.4$ Hz, 1H), 5.81 (dd, $J = 10.4, 1.5$ Hz, 1H), 4.06 (d, $J = 5.8$ Hz, 2H), 1.67 (t, $J = 6.0$ Hz, 1H), 1.36 – 1.24 (m, 16H), 0.94 – 0.84 (m, 6H)

C8: ^1H NMR (400 MHz, CDCl_3) δ : 6.38 (dd, $J = 16.9$ Hz $J = 1.6$ Hz, 1H); 6.11 (dd, $J = 10.4$ Hz $J = 16.9$ Hz, 1H); 5.80 (dd, $J = 10.4$ Hz $J = 1.6$ Hz, 1H); 4.05 (d, $J = 5.6$ Hz, 2H); 1.66 (m, 1H); 1.27 (m, 24H); 0.87 (t, $J = 6.7$ Hz, 3H)

C9: ^1H NMR (400 MHz, CDCl_3) δ : 6.39 (dd, $J = 17.3, 1.5$ Hz, 1H), 6.12 (dd, $J = 17.3, 10.4$ Hz, 1H), 5.81 (dd, $J = 10.4, 1.5$ Hz, 1H), 4.06 (d, $J = 5.8$ Hz, 2H), 1.58 (m, 1H), 1.36–1.23 (m, 30H), 0.92 – 0.84 (m, 6H).

Synthesis of lipid derivatives of A4B2C3 (Figure 3a)

The seven lipids were synthesized according to the general method describing in the article for the synthesis of STAAR lipids. Adding a final purification through flash chromatography with gradient elution from 100% of dichloromethane to 50% of 80/20/1 DCM/MeOH/ NH_4OH (aq).

The pure compounds were characterized by mass spectroscopy and ^1H NMR.

A4B2C3: (74 mg, 78%). Theoretical $[\text{M}+\text{H}]^+ = 684.57$, experimental $[\text{M}+\text{H}]^+ = 684.77$.

^1H NMR (400 MHz, CDCl_3) δ : 6.64 (m, 1H); 6.26 (d, $J = 8.0$ Hz, 1H); 4.60 (dt, $J = 7.0$ Hz $J = 7.0$ Hz, 1H); 3.99 (d, $J = 5.7$ Hz, 2H); 3.33 (dt, $J = 5.9$ Hz $J = 5.9$ Hz, 2H); 2.79 (t, $J = 7.5$ Hz, 2H); 2.58 (m, 4H); 2.40 (t, $J = 6.2$ Hz, 2H); 2.22 (s, 6H); 2.04 (m, 2H); 1.94 (m, 1H); 1.57 (m, 3H); 1.40 (m, 2H); 1.26 (m, 36H); 0.87 (m, 12H).

A4B2C8 (CP-LC-0729) (top-performing): (84 mg, 76%). Written in the article.

A4B2C9: (85 mg, 71%). Theoretical $[\text{M}+\text{H}]^+ = 796.69$, experimental $[\text{M}+\text{H}]^+ = 796.89$.

^1H NMR (400 MHz, CDCl_3) δ : 6.63 (t, $J = 4.8$ Hz, 1H); 6.26 (d, $J = 8.8$ Hz, 1H); 4.58 (dt, $J = 7.1$ Hz $J = 7.1$ Hz, 1H); 3.99 (d, $J = 5.6$ Hz, 2H); 3.32 (dt, $J = 5.8$ Hz $J = 5.8$ Hz, 2H); 2.79 (t, $J = 7.4$ Hz, 2H); 2.57 (m, 4H); 2.39 (t, $J = 6.0$ Hz, 2H); 2.21 (s, 6H); 2.04 (m, 2H); 1.96 (m, 1H); 1.58 (m, 3H); 1.41 (m, 2H); 1.25 (m, 52H); 0.87 (m, 12H).

A4B4C3: (74 mg, 77%). Theoretical $[\text{M}+\text{H}]^+ = 628.51$, experimental $[\text{M}+\text{H}]^+ = 628.65$.

^1H NMR (400 MHz, CDCl_3) δ : 6.69 (m, 1H); 6.32 (d, $J = 7.5$ Hz, 1H); 4.60 (dt, $J = 7.0$ Hz $J = 7.0$ Hz, 1H); 3.99 (d, $J = 6.0$ Hz, 2H); 3.33 (dt, $J = 6.0$ Hz $J = 6.0$ Hz, 2H); 2.79 (t, $J = 7.1$ Hz, 2H); 2.57 (m, 4H); 2.42 (t, $J = 6.1$ Hz, 2H); 2.23 (s, 6H); 2.05 (m, 2H); 1.95 (m, 1H); 1.58 (m, 3H); 1.41 (m, 2H); 1.25 (m, 28H); 0.87 (m, 12H).

A4B4C8: (75 mg, 73%). Theoretical $[M+H]^+=684.57$, experimental $[M+H]^+=684.75$.

^1H NMR (400 MHz, CDCl_3) δ : 6.67 (m, 1H); 6.28 (d, $J=8.0$ Hz, 1H); 4.59 (dt, $J=7.0$ Hz $J=7.0$ Hz, 1H); 3.99 (d, $J=6.0$ Hz, 2H); 3.33 (dt, $J=6.0$ Hz $J=6.0$ Hz, 1H); 2.79 (t, $J=7.4$ Hz, 2H); 2.58 (m, 4H); 2.21 (s, 6H); 2.05 (m, 2H); 1.94 (m, 1H); 1.59 (m, 3H); 1.41 (m, 2H); 1.25 (m, 36H); 0.87 (m, 12H).

A4B4C9: (90 mg, 81%). Theoretical $[M+H]^+=740.63$, experimental $[M+H]^+=740.83$.

^1H NMR (400 MHz, CDCl_3) δ : 6.65 (m, 1H); 6.26 (d, $J=7.8$ Hz, 1H); 4.59 (dt, $J=7.4$ Hz $J=7.4$ Hz, 1H); 3.99 (d, $J=5.8$ Hz, 2H); 3.32 (dt, $J=5.9$ Hz $J=5.9$ Hz, 2H); 2.79 (t, $J=7.2$ Hz, 2H); 2.59 (m, 4H); 2.40 (t, $J=6.1$ Hz, 2H); 2.22 (s, 6H); 2.05 (m, 2H); 1.95 (m, 1H); 1.58 (m, 3H); 1.41 (m, 2H); 1.25 (m, 60H); 0.87 (m, 12H).

A4B5C3: (85 mg, 77%). Theoretical $[M+H]^+=740.63$, experimental $[M+H]^+=740.86$.

^1H NMR (400 MHz, CDCl_3) δ : 6.64 (m, 1H); 6.27 (d, $J=8.0$ Hz, 1H); 4.60 (dt, $J=7.0$ Hz $J=7.0$ Hz, 1H); 3.99 (d, $J=5.8$ Hz, 2H); 3.33 (dt, $J=5.9$ Hz $J=5.9$ Hz, 2H); 2.79 (t, $J=7.4$ Hz, 2H); 2.58 (m, 4H); 2.40 (t, $J=6.1$ Hz, 2H); 2.22 (s, 6H); 2.04 (m, 2H); 1.95 (m, 1H); 1.56 (m, 3H); 1.41 (m, 2H); 1.25 (m, 44H); 0.87 (m, 12H).

A4B5C8: (89 mg, 75%). Theoretical $[M+H]^+=796.69$, experimental $[M+H]^+=796.87$.

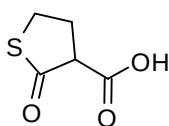
^1H NMR (400 MHz, CDCl_3) δ : 6.66 (t, $J=4.0$ Hz, 1H); 6.28 (d, $J=8.0$ Hz, 1H); 4.59 (dt, $J=7.2$ Hz $J=7.2$ Hz, 1H); 3.98 (d, $J=6.0$ Hz, 2H); 3.32 (dt, $J=5.7$ Hz $J=5.7$ Hz, 2H); 2.79 (t, $J=7.8$ Hz, 2H); 2.57 (m, 4H); 2.39 (t, $J=6.0$ Hz, 2H); 2.22 (s, 6H); 2.05 (m, 2H); 1.95 (m, 1H); 1.54 (m, 3H); 1.41 (m, 2H); 1.25 (m, 52H); 0.87 (m, 12H).

A4B5C9: (95 mg, 74%) Theoretical $[M+H]^+=852.76$, experimental $[M+H]^+=853.00$.

^1H NMR (400 MHz, CDCl_3) δ : 6.65 (m, 1H); 6.28 (d, $J=8.1$ Hz, 1H); 4.60 (dt, $J=7.2$ Hz $J=7.2$ Hz, 1H); 3.99 (d, $J=5.9$ Hz, 2H); 3.32 (dt, $J=5.8$ Hz $J=5.8$ Hz, 2H); 2.79 (t, $J=7.4$ Hz, 2H); 2.57 (m, 4H); 2.41 (t, $J=6.0$ Hz, 2H); 2.22 (s, 6H); 2.06 (m, 2H); 1.95 (m, 1H); 1.59 (m, 3H); 1.39 (m, 2H); 1.26 (m, 44H); 0.87 (m, 12H).

Preparation of intermediates required for the synthesis of derivatives of CP-LC-0729 (01 to 05, Figure 3b)

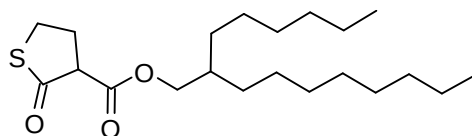
Synthesis of intermediate 2-oxotetrahydrothiophene-3-carboxylic acid.



The product was synthesized according to the literature¹.

^1H NMR (400 MHz, CD_3CN) δ : 2.55(dd, 1H), 3.50-3.46 (m, 2H), 2.43-2.61 (m, 2H).

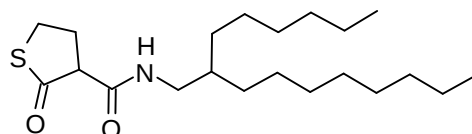
Synthesis of intermediate 2-hexyldecyl 2-oxotetrahydrothiophene-3-carboxylate.



2-Oxotetrahydrothiophene-3-carboxylic acid (175 mg, 1.2 mmol) was dissolved in 5 mL of anhydrous dichloromethane at room temperature. Then 2-hexyl-1-decanol (299 mg, 1.0 mmol) was added followed by EDC hydrochloride (254 mg, 1.3 mmol) and 4-(dimethylamino)pyridine (25 mg, 0.2 mmol). The reaction mixture was stirred at room temperature overnight under an argon atmosphere. After that, the reaction mixture was first washed twice with water and finally with brine. The organic layer was dried with anhydrous MgSO_4 , filtered, and evaporated under reduced pressure. The resulting residue was purified by flash chromatography (gradient of hexane/dichloromethane: 100/0 to 0/100) to afford the product as a colorless liquid (271 mg, 73%).

^1H NMR (400 MHz, CDCl_3) δ : 4.15 – 4.00 (m, 2H), 3.57 – 3.45 (m, 2H), 3.36 (m, 1H), 2.64 (m, 1H), 2.50 (m, 1H), 1.70 – 1.62 (m, 1H), 1.27 (m, 24H), 0.92 – 0.84 (m, 6H).

Synthesis of intermediate N-(2-hexyldecyl)-2-oxotetrahydrothiophene-3-carboxamide.

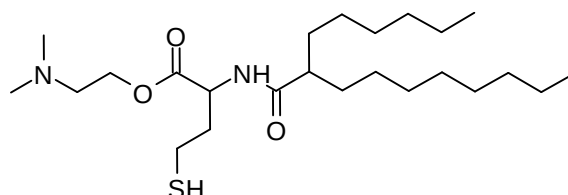


2-Oxotetrahydrothiophene-3-carboxylic acid (175 mg, 1.2 mmol) was dissolved in 5 mL of anhydrous dichloromethane at room temperature. Then 2-hexyldecyl-1-amine (170 mg, 1.0 mmol) was added followed by EDC hydrochloride (254 mg, 1.3 mmol) and 4-(dimethylamino)pyridine (25 mg, 0.2 mmol). The reaction mixture was stirred at room temperature overnight under an argon atmosphere. After that, the reaction mixture was first washed twice with water and finally with brine. The organic layer was dried with anhydrous MgSO_4 , filtered, and evaporated under reduced pressure. The resulting residue was purified by flash chromatography (gradient of hexane/ethyl acetate: 100/0 to 0/100) to afford the product as a whitish solid (248 mg, 67%).

The pure compound was characterized by mass spectroscopy.

Theoretical $[\text{M}+\text{H}]^+ = 370.28$, experimental $[\text{M}+\text{H}]^+ = 370.46$

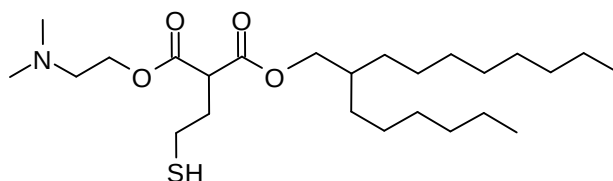
2-(Dimethylamino)ethyl (2-hexyldecanyl)homocysteinate.



(2-Hexyldecanoyl)homocysteine thiolactone (**B2**) (160 mg ,0.36 mmol) was added over 2-(dimethylamino)ethanol (2.00 ml, 20 mmol), the reaction mixture was heated at 85°C for 16 hours under argon atmosphere. Subsequently, the reaction mixture was first washed with 1M HCl, once with water and finally with brine. The organic layer was dried with anhydrous MgSO₄, filtered, and evaporated under reduced pressure. The resultant residue was a mixture of B2 (45%) and the target product (55%). This residue was used in the next step without further purification.

The pure compound was characterized by mass spectroscopy. Theoretical [M+H]⁺=445.35, experimental [M+H]⁺=445.52

1-(2-(Dimethylamino)ethyl) 3-(2-hexyldeacyl) 2-(2-mercaptoethyl)malonate.



2-Hexyldeacyl 2-oxotetrahydrothiophene-3-carboxylate (173 mg ,0.47 mmol) was added in 2-(dimethylamino)ethanol (2.59 ml, 26 mmol), the reaction mixture was heated at 85°C during 16 hours under argon atmosphere. Subsequently, the reaction mixture was first washed with 1M HCl, once with water and finally with brine. The organic layer was dried with anhydrous MgSO₄, filtered, and evaporated under reduced pressure. The resultant residue was a mixture of 1-(2-(dimethylamino)ethyl) 3-(2-hexyldeacyl) 2-(2-mercaptoethyl)malonate (50%) and the target product (50%). This residue was used in the next step without further purification (91 mg, 43%).

The pure compound was characterized by mass spectroscopy. Theoretical [M+H]⁺=460.35, experimental [M+H]⁺=460.52

Synthesis lipid derivatives of CP-LC-0729 (01 to 05, Figure 3b).

Lipid CP-LC-0729 and three of its variants (CP-LC-0729-02, CP-LC-0729-03 and CP-LC-0729-04) were synthesized according to the general method described in the article for the synthesis of STAAR lipids using their corresponding initial products. Adding a final purification through flash chromatography with gradient elution from 100% of dichloromethane to 50% of 80/20/1 DCM/MeOH/ NH₄OH (aq) allowed the obtention of these products in the following purities and yields:

A4B2C8 (CP-LC-0729) (top-performing): (84 mg, 76%). Written in the article.

Theoretical [M+H]⁺=740.63, experimental [M+H]⁺=740.84.

¹H NMR (400 MHz, CDCl₃) δ: 6.76 (m, 1H); 6.37 (d, J= 7.9 Hz, 1H); 4.60 (dt, J= 7.1 Hz J= 7.1 Hz , 1H); 3.98 (d, J= 5.8 Hz, 2H); 3.36 (dt, J= 5.7 Hz J= 5.7 Hz, 2H); 2.78 (t, J= 6.8 Hz, 2H); 2.59 (m, 4H); 2.48 (t, J= 5.9 Hz, 2H); 2.28 (s, 6H); 2.06 (m, 2H); 1.96 (m, 1H); 1.58 (m, 3H); 1.41 (m, 2H); 1.25 (m, 44H); 0.87 (m, 12H).

CP-LC-0729-02: A3, B2 and C8 were used as initial products. (85 mg, 75%).

Theoretical $[M+H]^+=754.65$, experimental $[M+H]^+=754.80$.

^1H NMR (400 MHz, CDCl_3) δ : (400 MHz, CDCl_3) δ : *6.41 (d, $J=8.0$ Hz, 0.66H); *6.26 (d, $J=8.0$ Hz, 0.34H); 5.06 (m, 1H); 3.98 (d, $J=5.6$ Hz, 2H); 3.49 (m, 2H); 3.12 (s, 2H); 2.96 (s, 1H); 2.77 (m, 2H); 2.56 (m, 4H); 2.44 (m, 2H); 2.28 (s, 2H); 2.24 (s, 4H); 2.01 (m, 2H); 1.84 (m, 1H); 1.59 (m, 3H); 1.39 (m, 1H); 1.26 (m, 44H); 0.87 (m, 12H). N-methylamide group has partial double-bond character (resonance between $\text{C}=\text{O}$ and C-N). This rigidity can affect the coupling constants and relative chemical shifts of protons nearby. * proton integration of the amide group evidences the rigidity of this bond.

CP-LC-0729-03: A4, C8 and N-(2-hexyldecyl)-2-oxotetrahydrothiophene-3-carboxamide were used as initial products. (79 mg, 70%).

Theoretical $[M+H]^+=754.24$, experimental $[M+H]^+=754.81$.

^1H NMR (400 MHz, CDCl_3) δ : 6.88 (m, 2H); 3.99 (d, $J=5.8$ Hz, 2H); 3.33 (dt, $J=5.8$ Hz, $J=5.8$ Hz, 2H); 3.18 (m, 3H); 2.77 (t, $J=7.4$ Hz, 2H); 2.57 (m, 4H); 2.42 (t, $J=6.2$ Hz, 2H); 2.23 (s, 6H); 2.13 (m, 2H); 1.62 (m, 1H); 1.48 (m, 1H); 1.26 (m, 48H); 0.88 (m, 12H).

CP-LC-0729-04: A4, C8 and 2-hexyldecyl 2-oxotetrahydrothiophene-3-carboxylate were used as initial products. (69 mg, 61%).

Theoretical $[M+H]^+=755.63$, experimental $[M+H]^+=755.83$.

^1H NMR (400 MHz, CDCl_3) δ : 6.79 (m, 1H); 4.05 (d, $J=5.8$ Hz, 2H); 4.01 (d, $J=5.8$ Hz, 2H); 3.45 (t, $J=7.1$ Hz, 1H); 3.36 (m, 2H); 2.79 (t, $J=7.6$ Hz, 2H); 2.59 (m, 4H); 2.44 (t, $J=6.0$ Hz, 2H); 2.26 (s, 6H); 2.21 (m, 2H); 1.65 (m, 2H); 1.29 (m, 48H); 0.90 (m, 12H).

The other two lipids required of alternative synthetic methods as follows:

CP-LC-0729-01: Intermediate 2-(dimethylamino)ethyl (2-hexyldecanoyl)homocysteinate (89 mg, 0.2 mmol) (described in the previous section) and acrylate C8 (89 mg, 0.3 mmol) were dissolved in 0.4 ml tetrahydrofuran. Then, triethylamine was added (31 mg, 0.3 mmol). The reaction mixture was stirred at room temperature for 16 hours under argon atmosphere. Later, the crude was purified through flash chromatography with gradient elution from 100% of dichloromethane to 50% of 80/20/1 DCM/MeOH/ NH_4OH (aq) (62 mg, 42%).

Theoretical $[M+H]^+=741.62$, experimental $[M+H]^+=741.84$.

^1H NMR (400 MHz, CDCl_3) δ : 6.29 (d, $J=7.8$ Hz, 1H); 4.73 (dt, $J=5.1$, $J=7.5$ Hz, 1H); 4.25 (td, $J=5.6$, 2.1 Hz, 2H); 3.99 (d, $J=5.8$ Hz, 2H); 2.76 (m, 2H); 2.59 (m, 6H); 2.29 (s, 6H); 2.09 (m, 3H); 1.59 (m, 3H); 1.42 (m, 2H); 1.26 (m, 44H); 0.87 (m, 12H).

CP-LC-0729-05: Intermediate 1-(2-(dimethylamino)ethyl) 3-(2-hexyldecyl) 2-(2-mercaptoethyl)malonate (91 mg, 0.2 mmol) (described in the previous section) and C8

(119mg, 0.4 mmol) were dissolved in 0.4 ml tetrahydrofuran. Then, triethylamine was added (56 mg, 0.4 mmol). The reaction mixture was stirring at room temperature during 16 hours under argon atmosphere. Later, the crude was purified through flash chromatography with gradient elution from 100% of dichloromethane to 50% of 80/20/1 DCM/MeOH/ NH₄OH (aq). (56 mg, 37%)

Theoretical [M+H]⁺=756.62, experimental [M+H]⁺=756.78.

¹H NMR (400 MHz, CDCl₃) δ: 4.25 (m, 2H); 4.05 (d, J= 5.8 Hz, 2H); 4.00 (d, J= 5.8 Hz, 2H); 3.59 (t, J= 7.2 Hz, 1H); 2.77 (t, J= 7.4 Hz, 2H); 2.58 (m, 6H); 2.28 (s, 6H); 2.21 (m, 2H); 1.62 (m, 2H); 1.27 (m, 48H); 0.90 (m, 12H).

Synthesis cationic lipids

(+) **A2B2C4**: Lipid A2B2C4 (149.5 mg, 0.20 mmol) was dissolved in toluene (0.7 ml) under Argon atmosphere. Then, dimethylsulfate (22.9 ul, 0.24 mmol) was added and one hour later the solvent was evaporated under reduced pressure. Afterwards, the crude reaction was purified using a CombiFlash NextGen 300+ with gradient elution from 100% of DCM to 40% of MeOH to afford the pure product (63%).

MS-QDa: theoretical [M]⁺ = 754.65, experimental [M]⁺ = 754.60

¹H NMR (400 MHz, CD₃CN) δ: 7.37 (m, 1H); 7.22 (d, J= 7.4 Hz, 1H); 4.36 (m, 2H); 4.03 (d, J= 6.6 Hz, 2H); 3.52 (m, 5H); 3.28 (m, 6H); 2.92 (s, 3H); 2.73 (t, J= 7.1 Hz, 2H); 2.55 (m, 4H); 2.19 (m, 1H); 1.88 (m, 2H); 1.58 (m, 2H); 1.49 (m, 2H); 1.27 (m, 50H); 0.87 (m, 9H).

(+) **A4B2C1**: Lipid A4B2C1 (116.1 mg, 0.15 mmol) was dissolved in toluene (0.5 ml) under Argon atmosphere. Then, dimethylsulfate (17.2 ul, 0.18 mmol) was added and one hour later the solvent was evaporated under reduced pressure. Afterwards, the crude reaction was purified using a CombiFlash NextGen 300+ with gradient elution from 100% of DCM to 40% of MeOH to afford the pure product (65%).

MS-QDa: theoretical [M]⁺ = 780.66, experimental [M]⁺ = 780.60.

¹H NMR (400 MHz, CD₃CN) δ: 7.43 (m, 1H), 7.03 (d, J = 7.8 Hz, 1H), 5.36 (m, 2H), 4.36 (m, 1H), 4.04 (t, J = 6.7 Hz, 2H), 3.55 (m, 5H), 3.39 (m, 2H), 3.07 (s, 9H), 2.73 (m, 2H), 2.59 (m, 4H), 2.19 (m, 1H), 2.01 (m, 4H), 1.86 (m, 1H), 1.58 (m, 2H), 1.48 (m, 2H), 1.28 (m, 44H), 0.88 (m, 9H).

SUPPORTING FIGURES AND TABLES

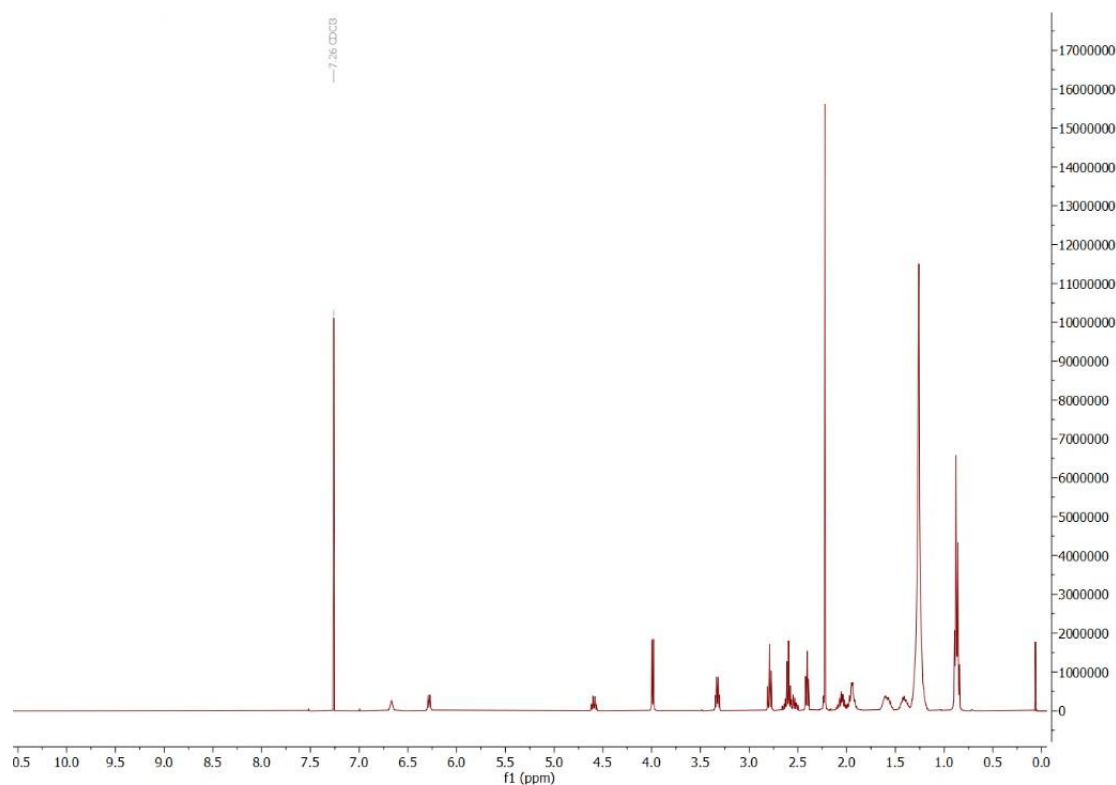
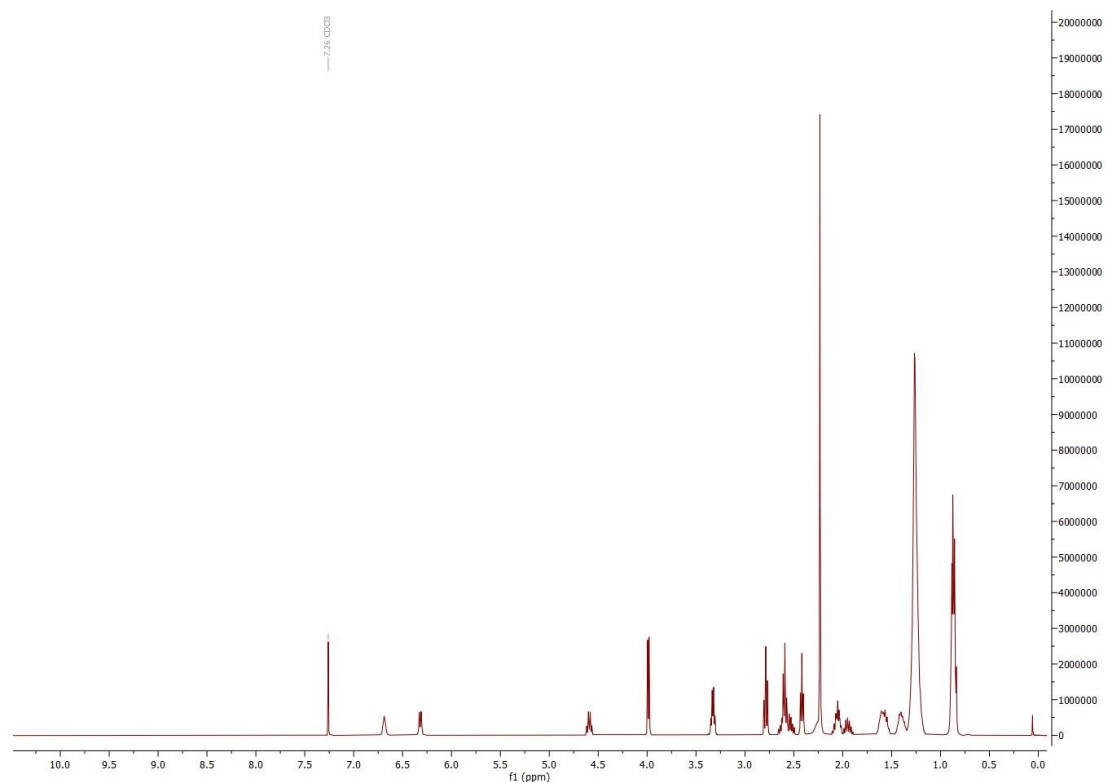
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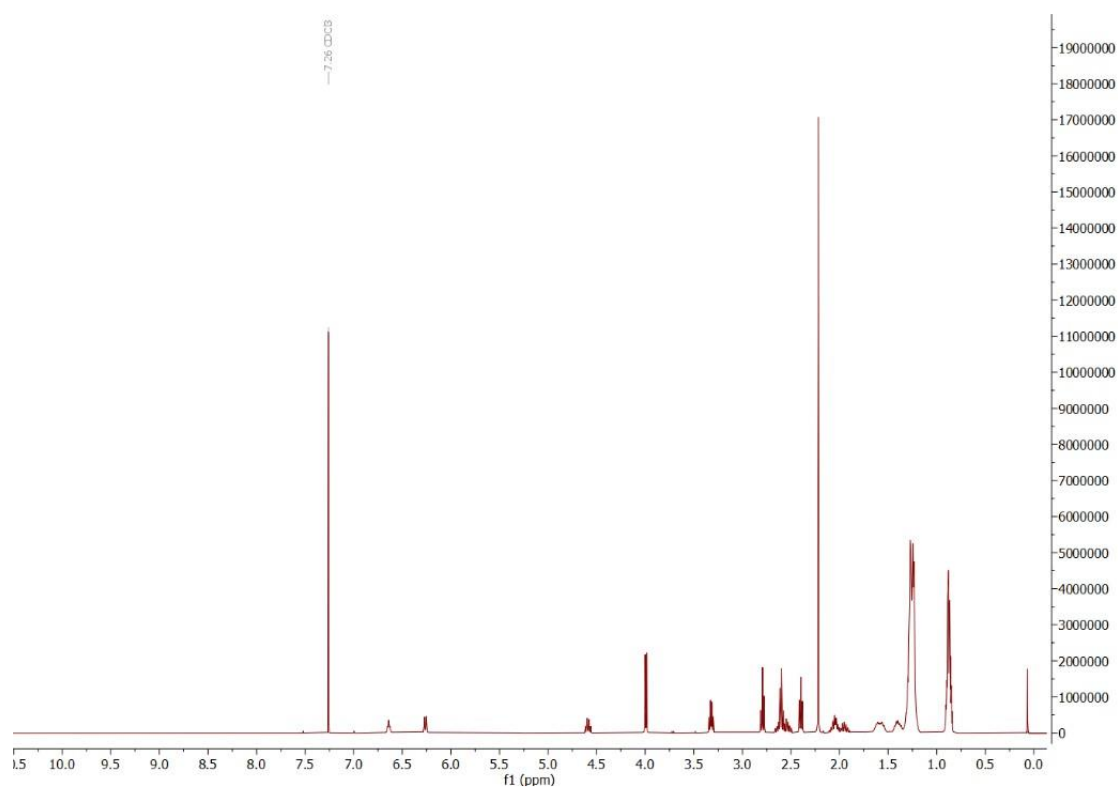
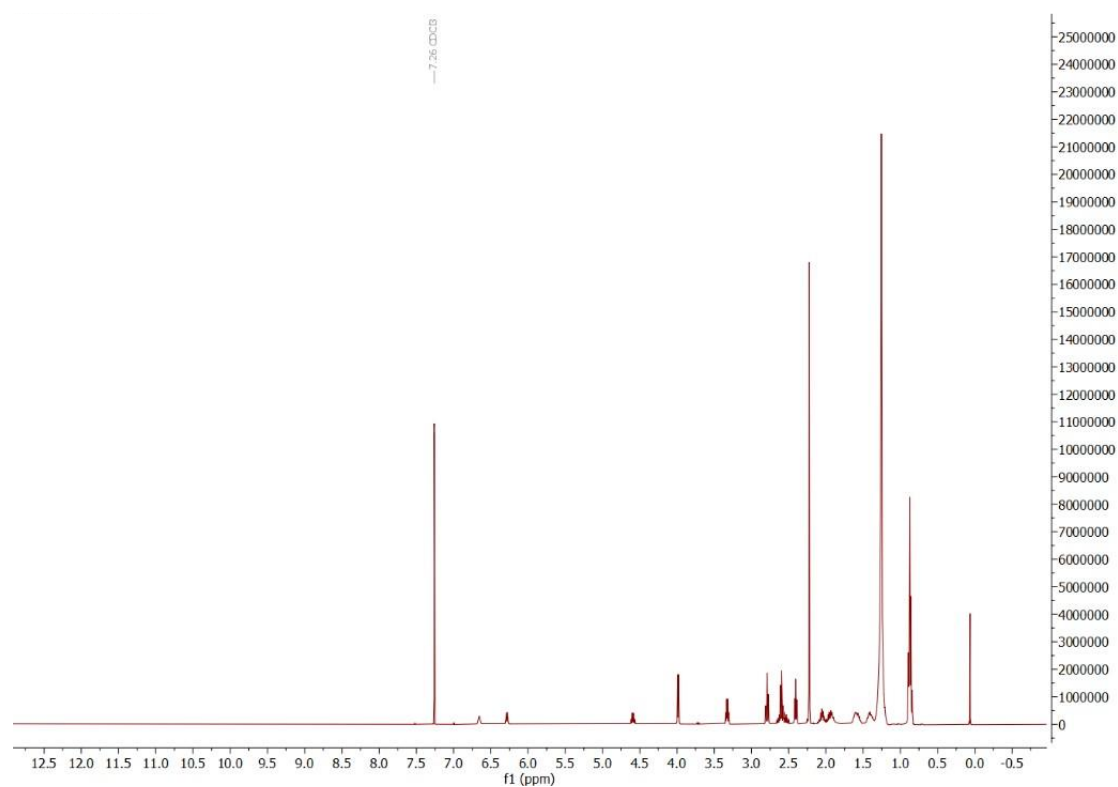
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AA
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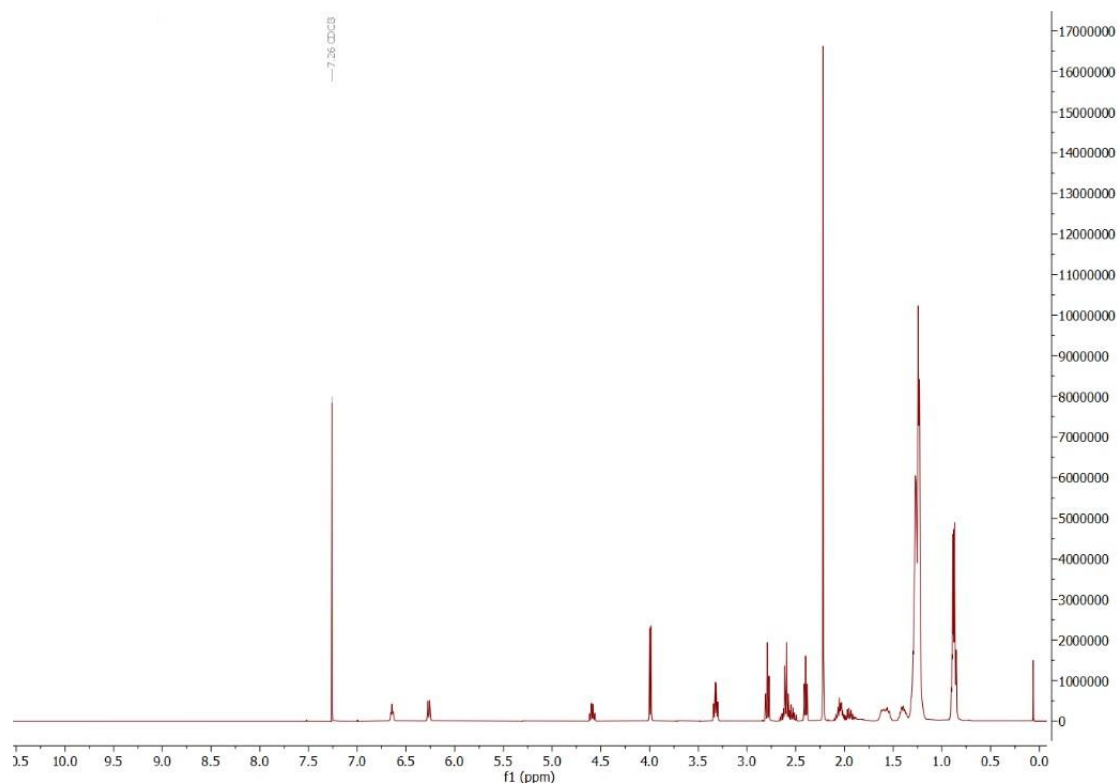
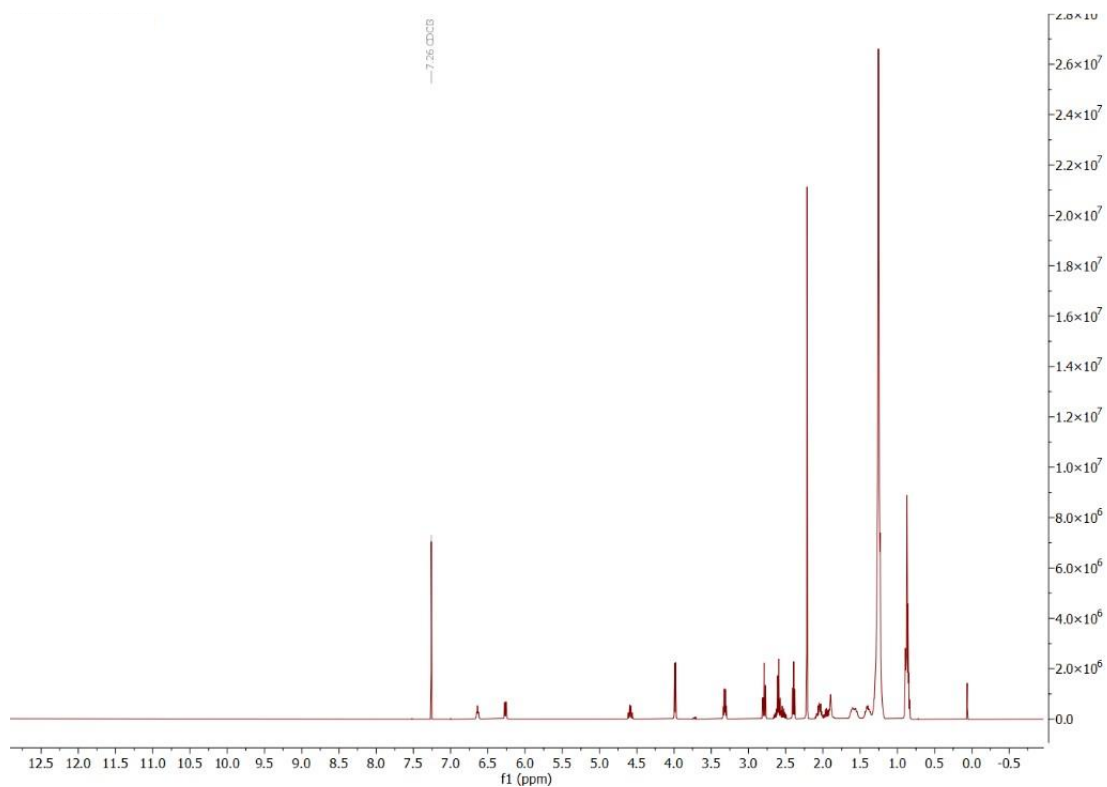
Figure S1. Firefly luciferase mRNA complete sequence. Where Ψ= 1-methyl-3'-pseudouridylyl^{2,3}.

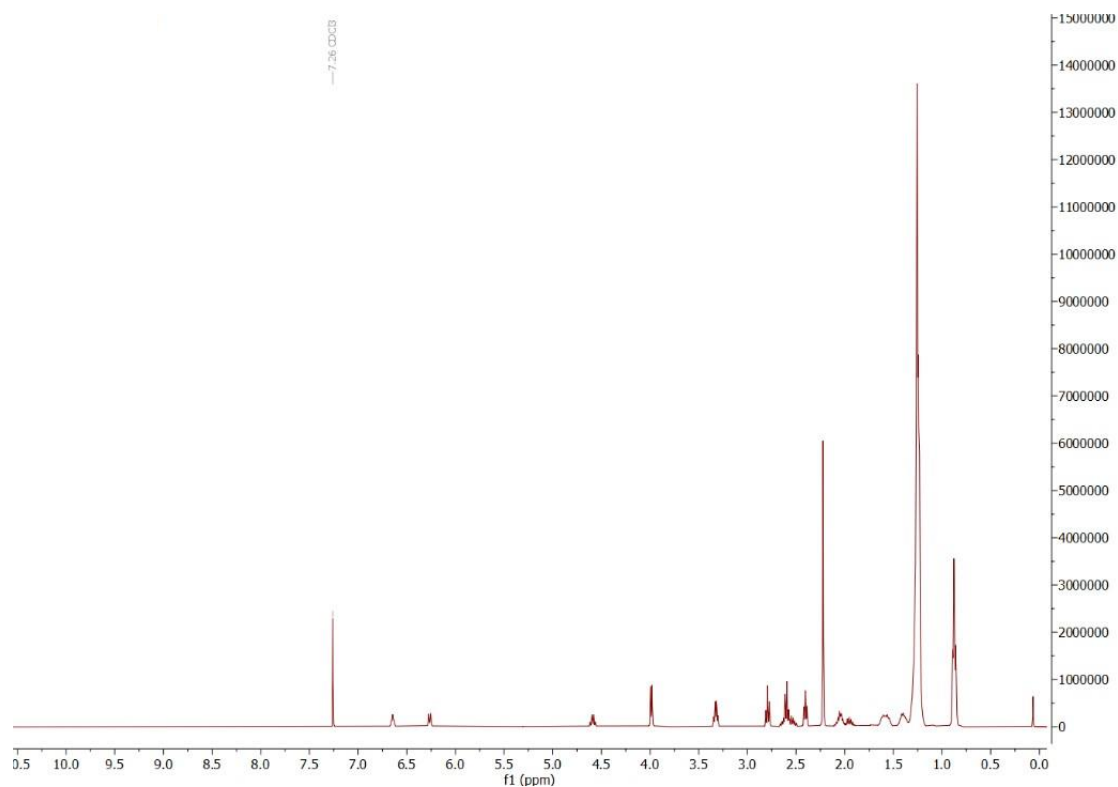
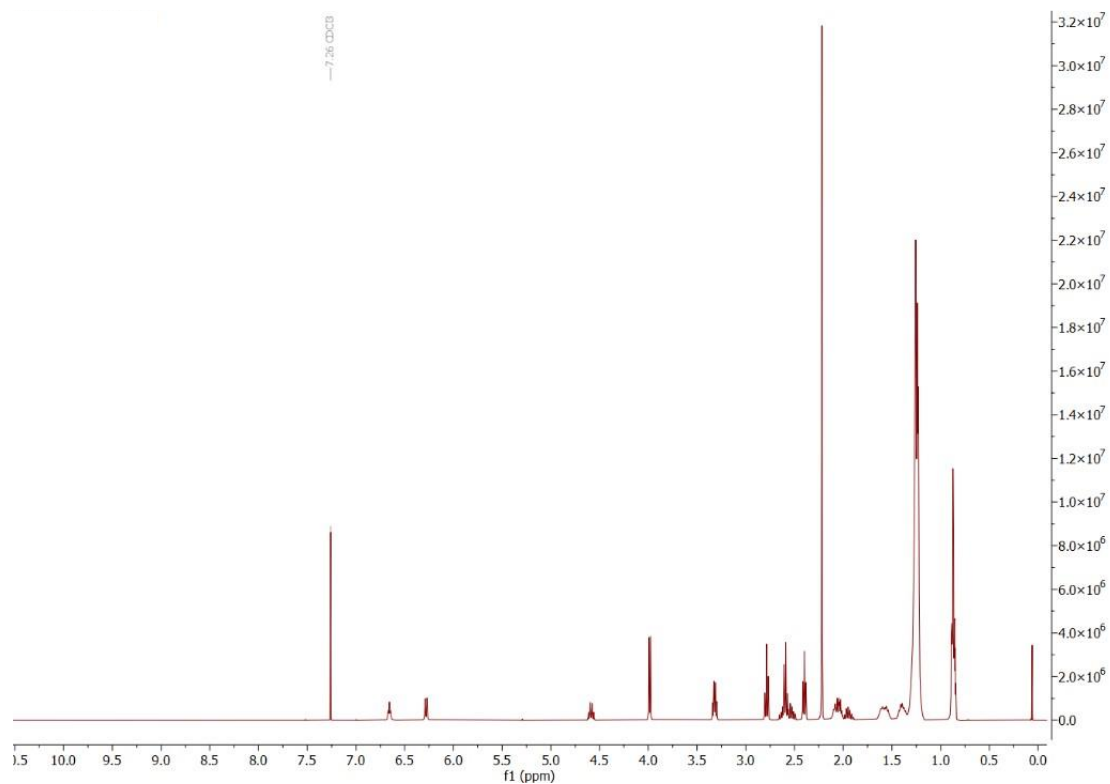
NMR spectra

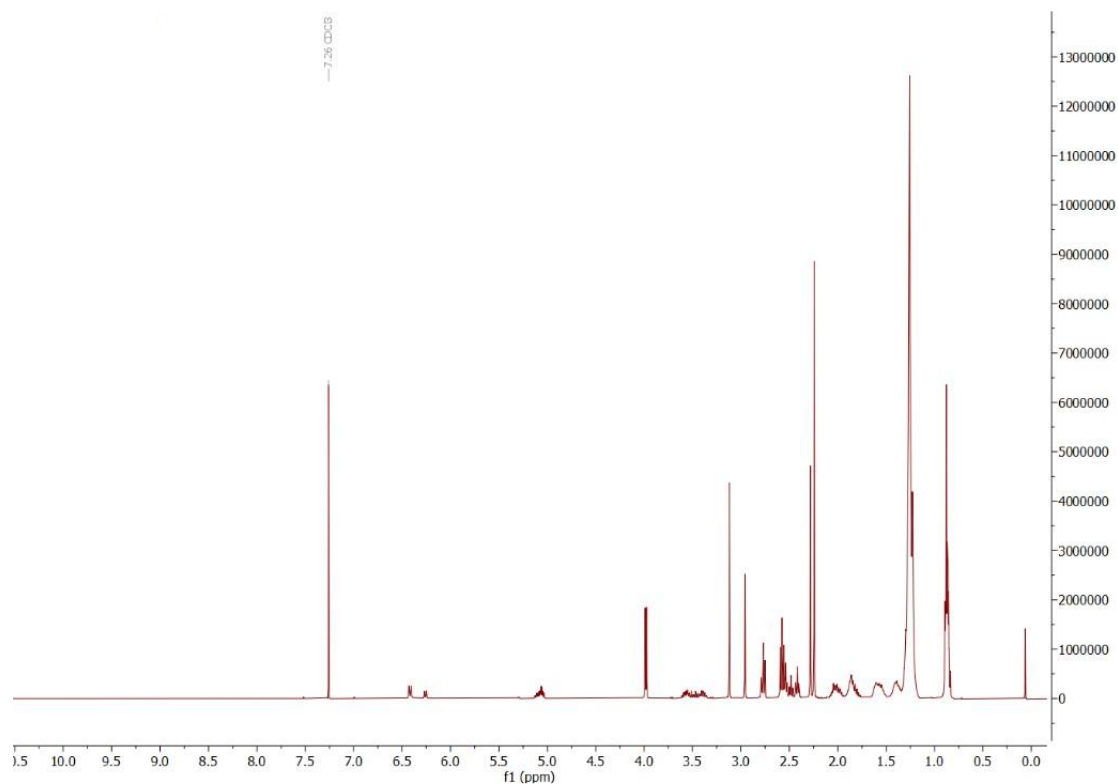
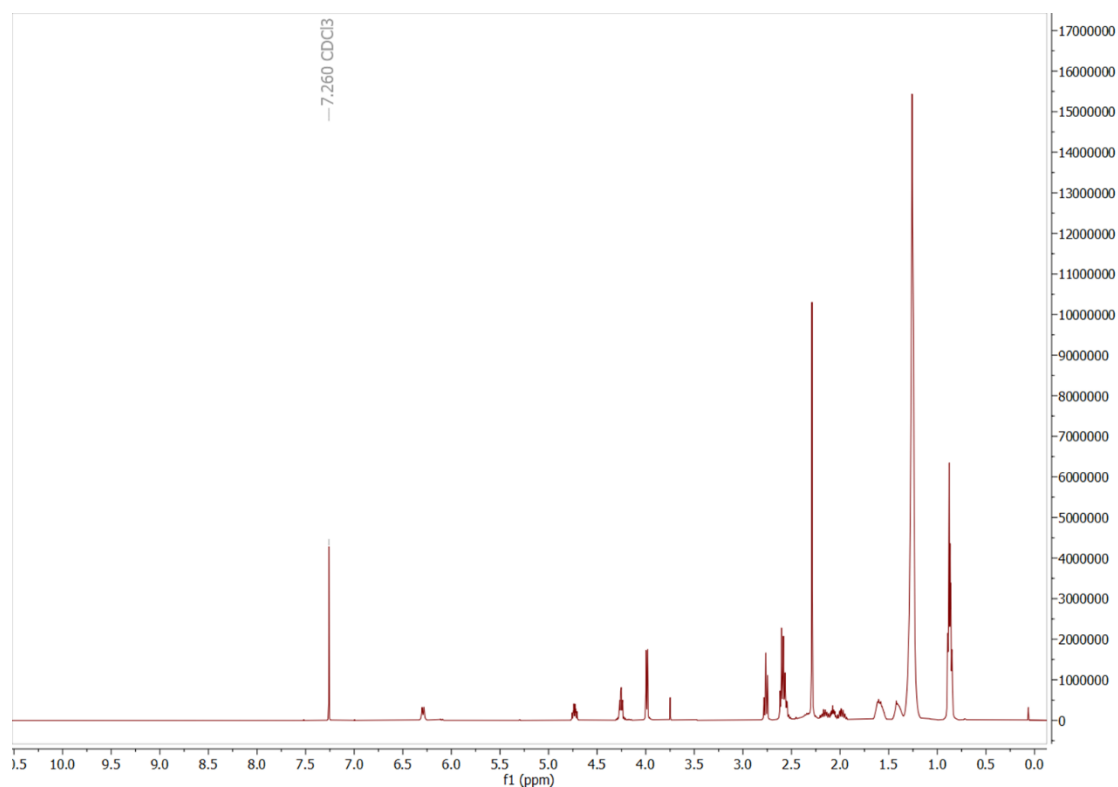
All NMR spectra were obtained using a Bruker Avance 400 MHz device and deuterated solvents (CDCl₃, Sigma Aldrich; DMSO-d₆, Sigma Aldrich; CD₃CN, Sigma Aldrich). All CDCl₃ spectra contains approximately 0.03% of tetramethyl silane (TMS).

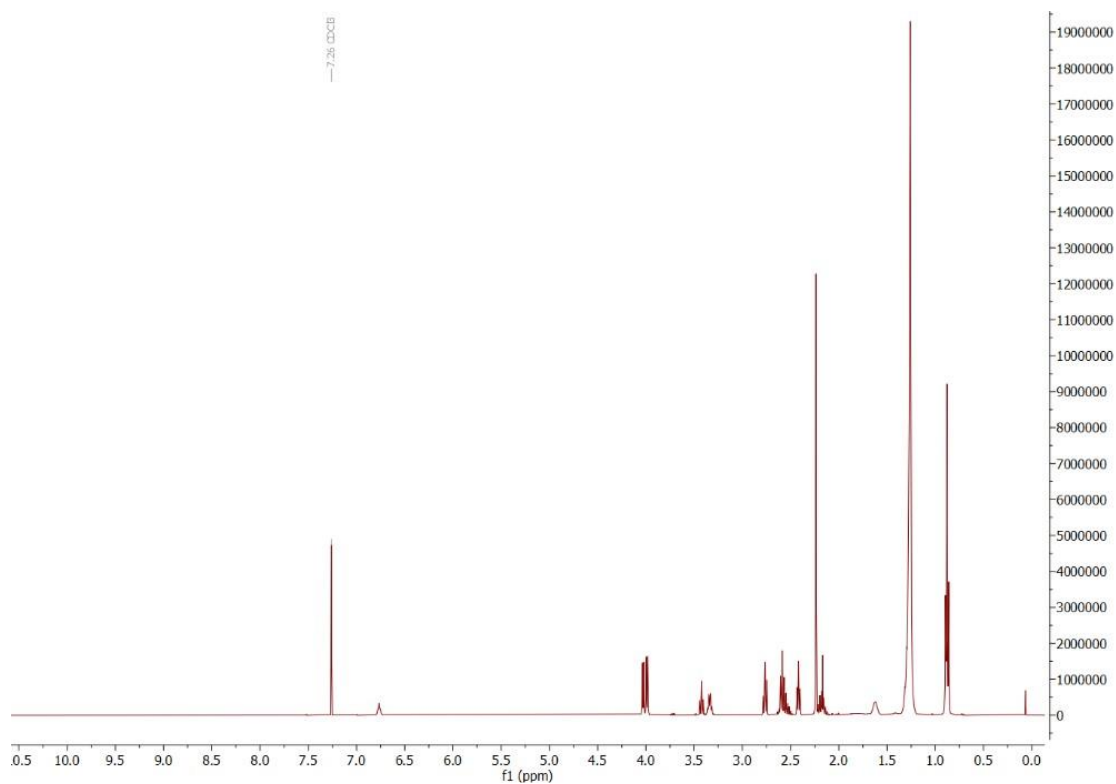
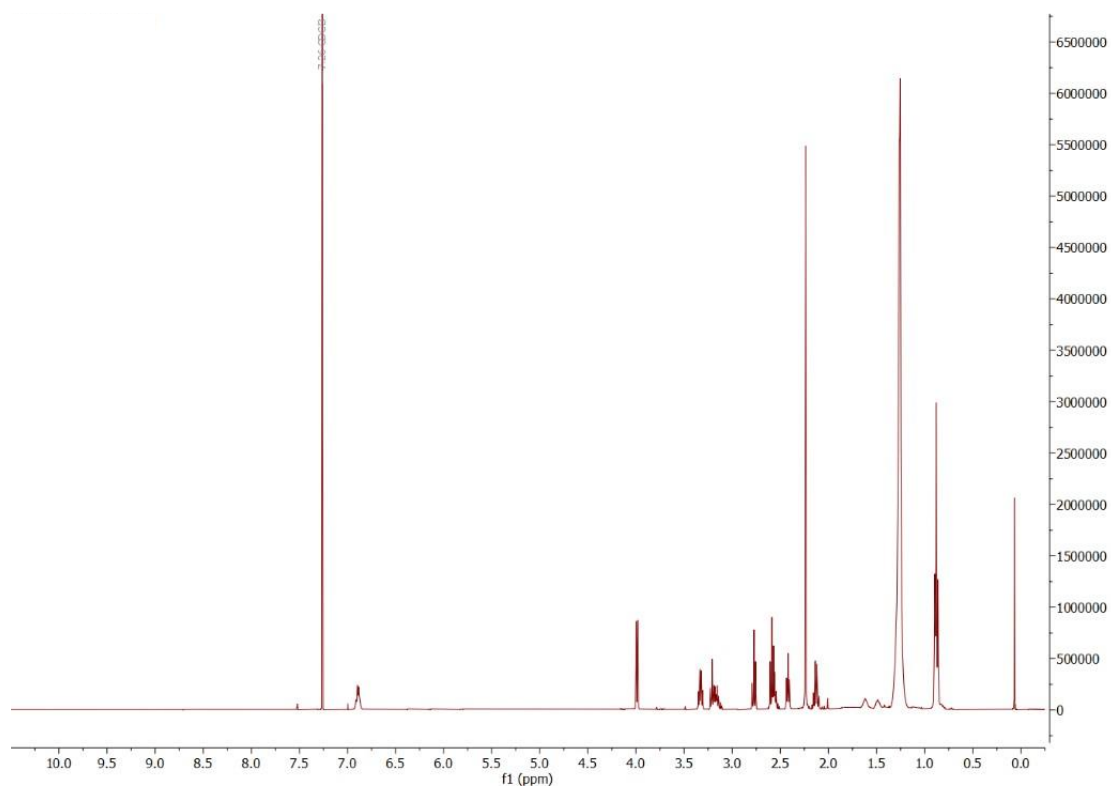


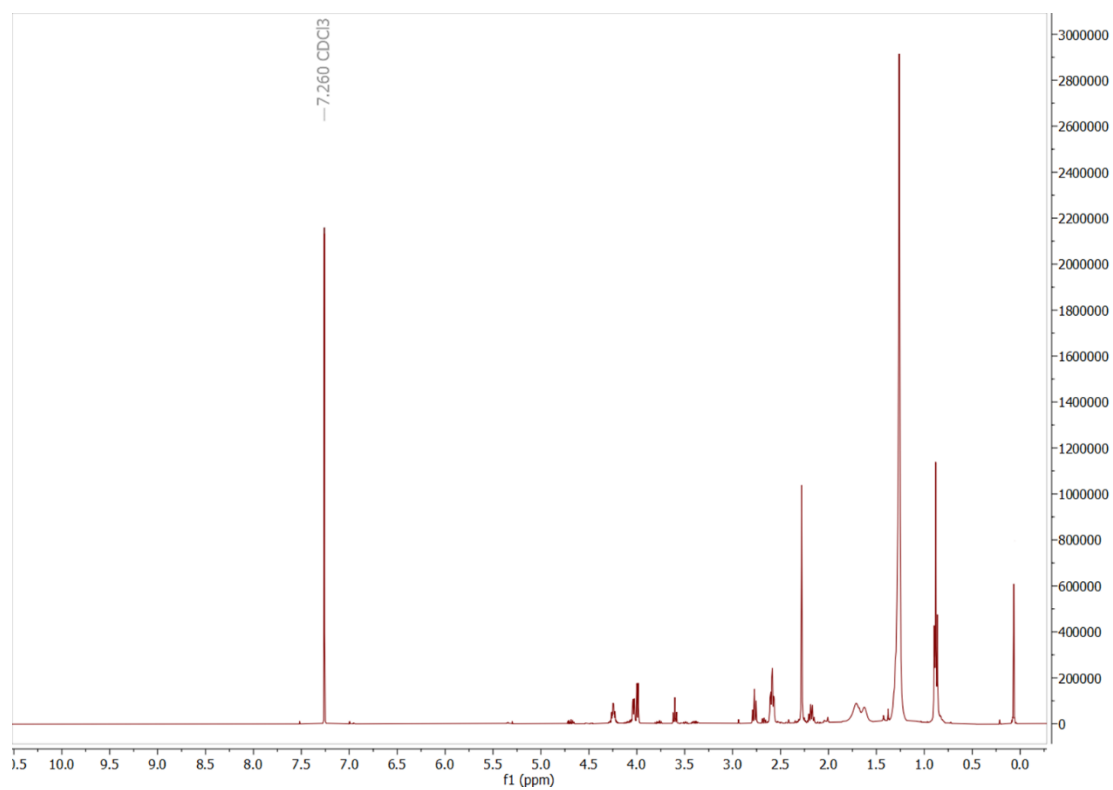




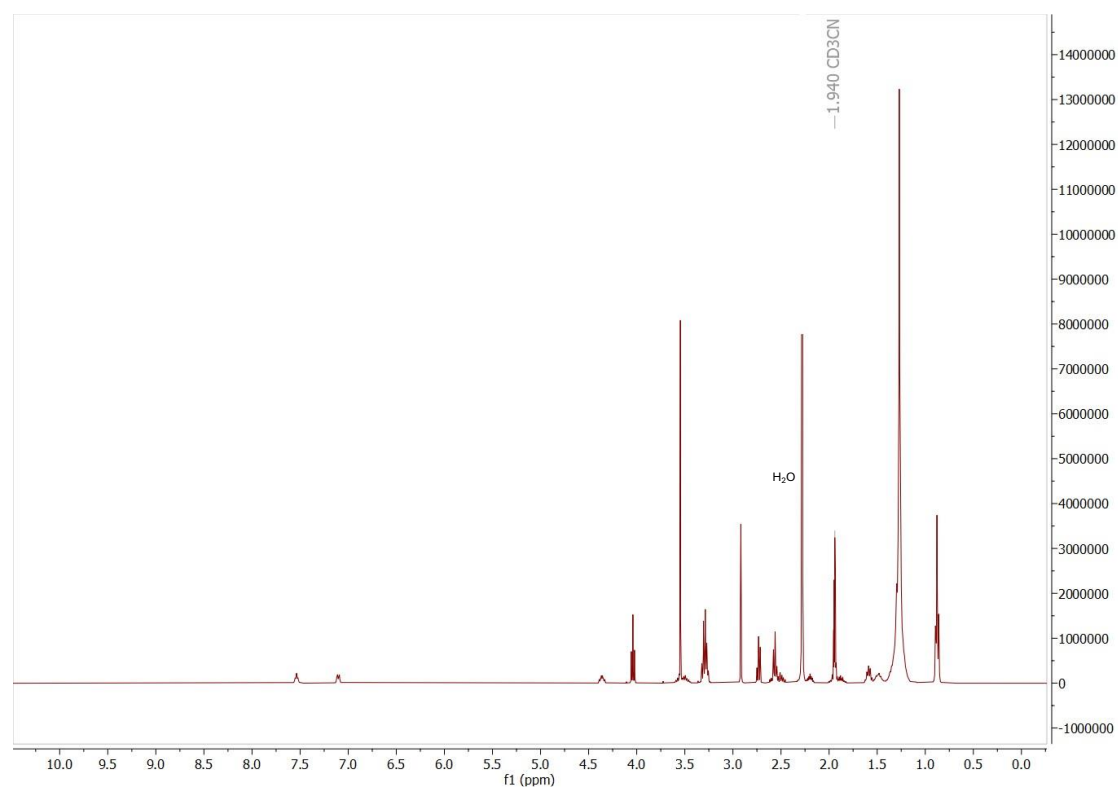




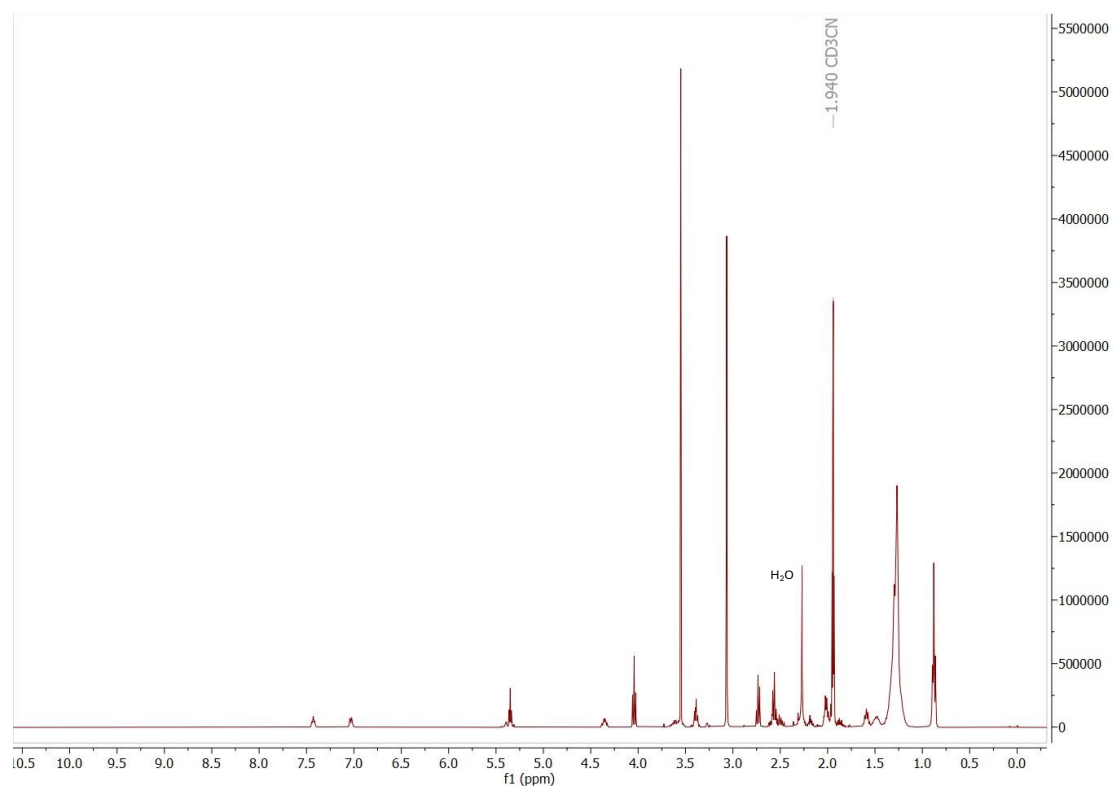




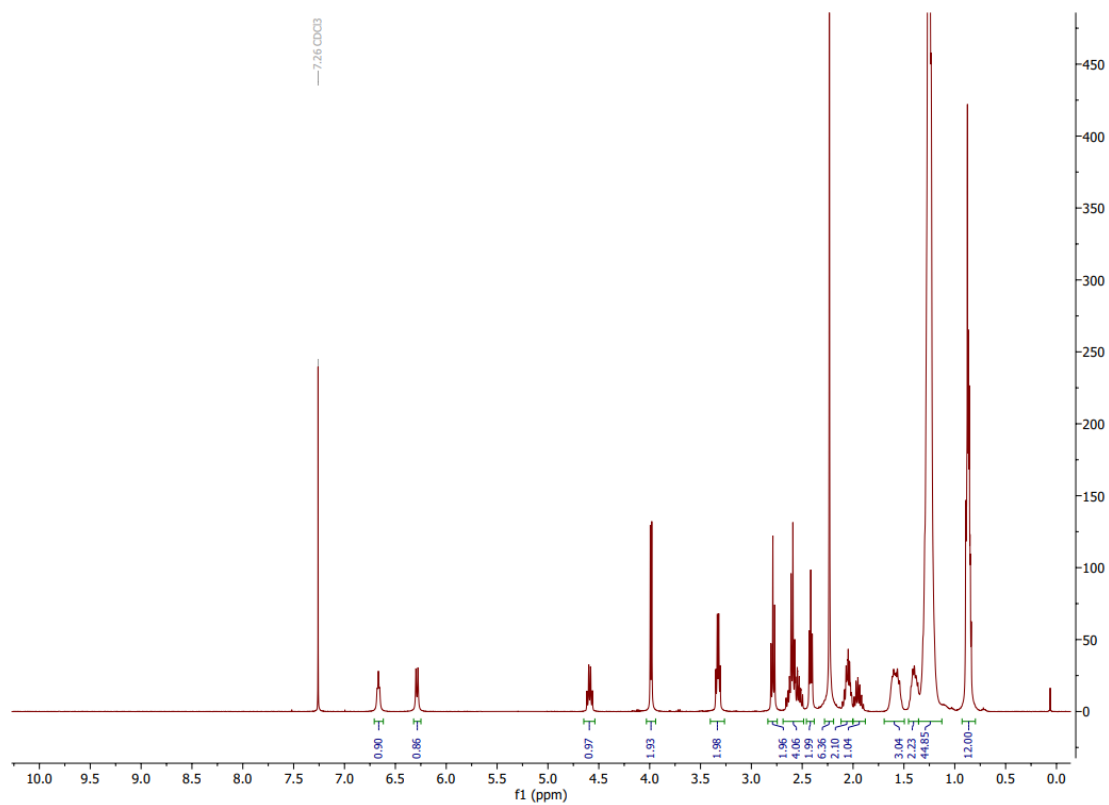
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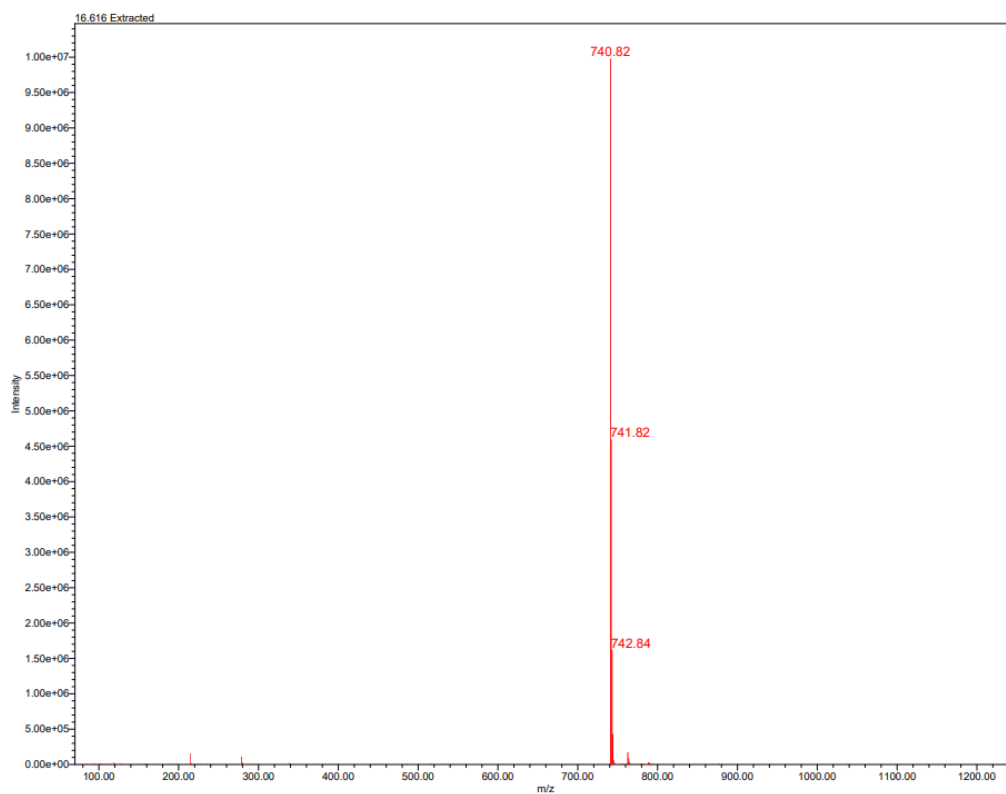
^1H NMR (400 MHz, CD_3CN) (+) A4B2C4



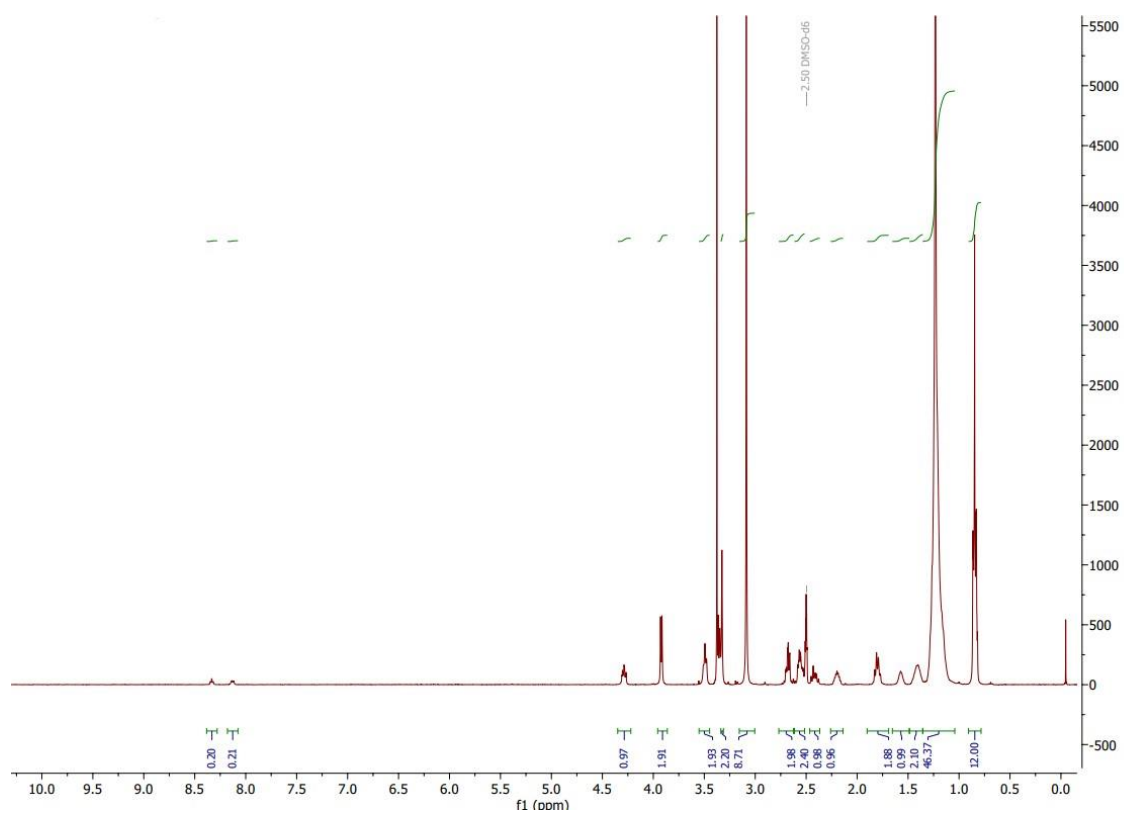
^1H NMR (400 MHz, CD_3CN) (+) A4B2C1



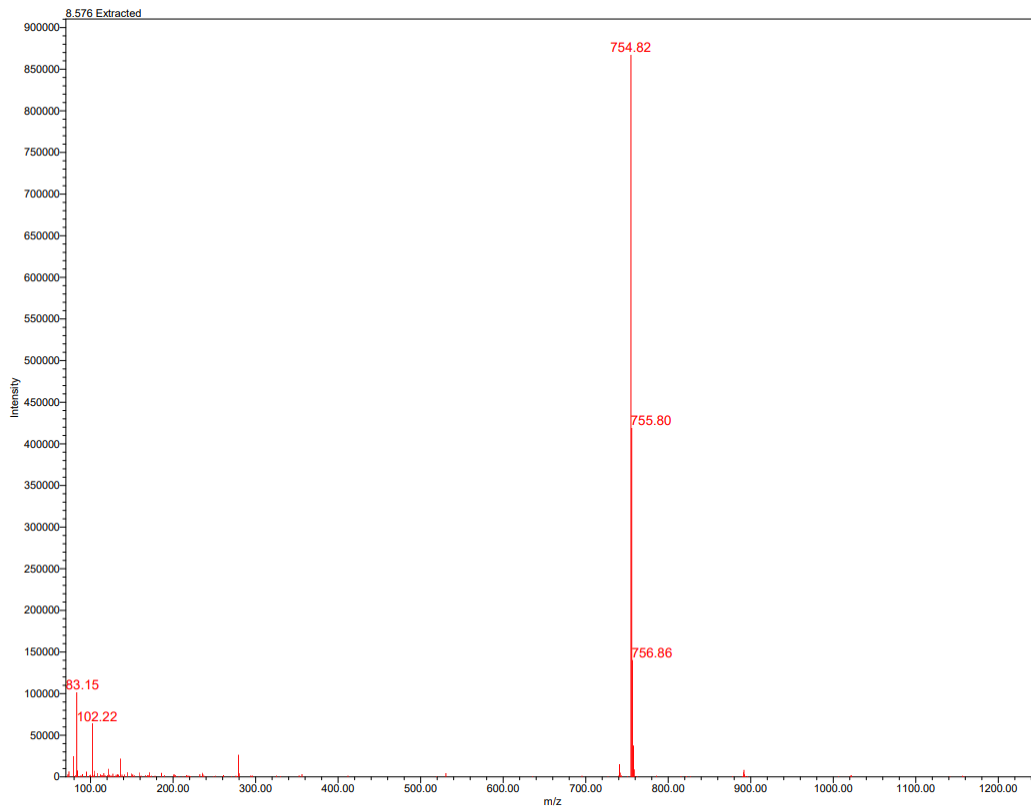
¹H NMR (400 MHz, CDCl₃) CP-LC-0729 (A4B2C8)



Mass Spectrometry Spectrum CP-LC-0729 (A4B2C8)



¹H NMR (400 MHz, DMSO-d₆) (+) CP-LC-0729



Mass Spectroscopy Spectrum CP-LC-0729 (A4B2C8)

Figure S2. ^1H NMR spectra of purified lipids (ionizables and cationics) along with mass spectrum and ^1H NMR of purified CP-LC-0729 and (+) CP-LC-0729.

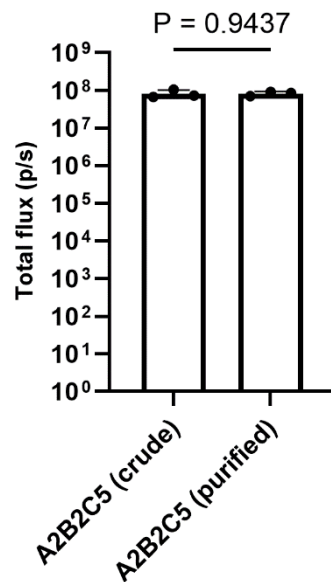


Figure S3. Comparison of purified and crude LNPs of lipid A2B2C5 (in vivo expression). In vivo mLuc expression (n = 3 biologically independent samples). Data are presented as mean values $\pm$ SD. Mice were i.m. injected with mLuc-loaded LNPs at an

mRNA dose of 0.05 mg/kg. Luminescence imaging was performed at 4 h post-treatment and total flux was quantified. Student's t-test was used.

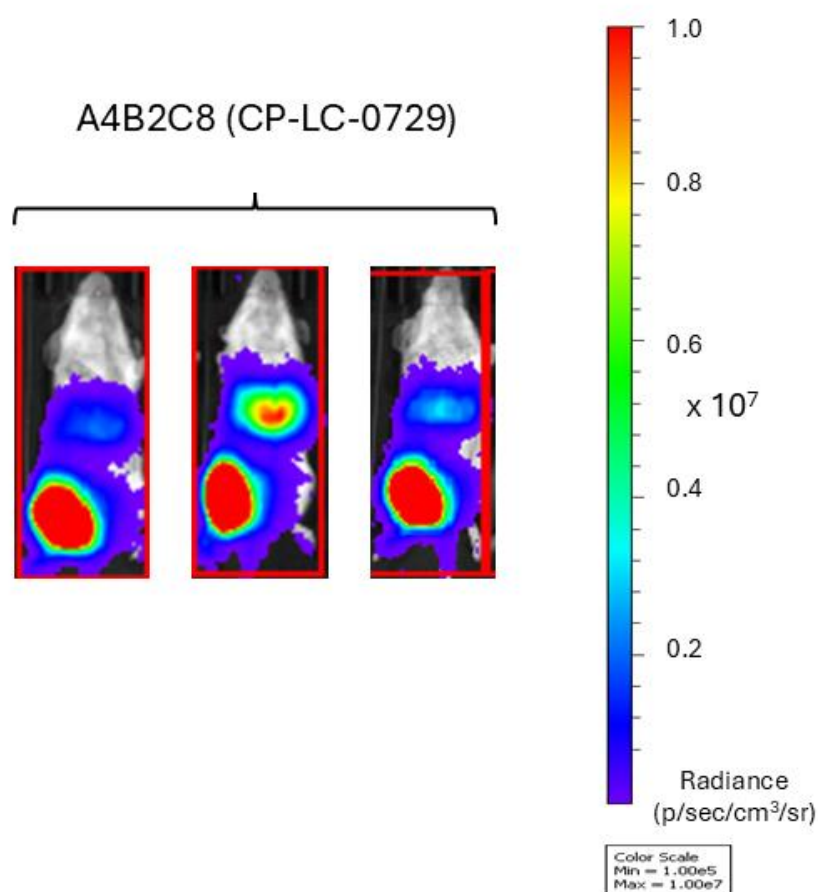


Figure S4. Image of mice inoculated intramuscularly with LNPs containing CP-LC-0729 (n=3) after 4h, using luciferase mRNA and the standard formulation (see Methods).

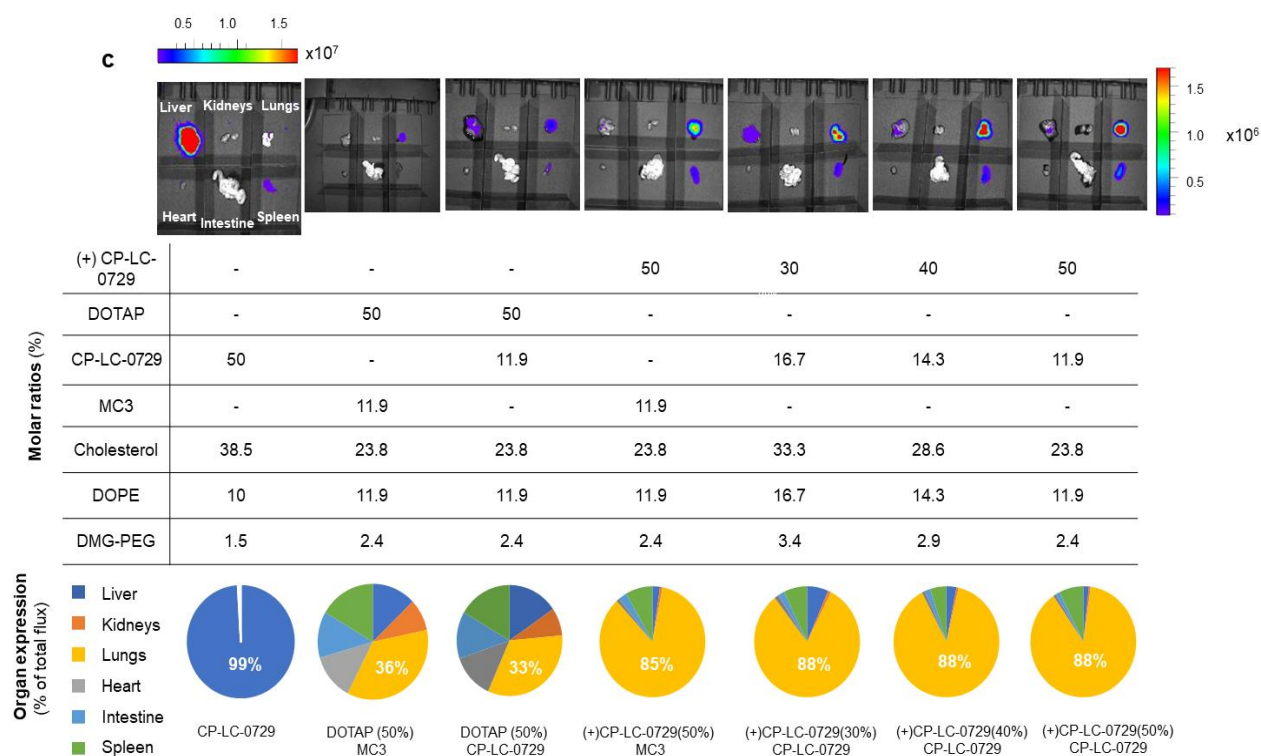


Figure S5. *Ex vivo* luminescence of major organs from mice and their quantification (n= 3 biologically independent samples). Mice were i.v. injected with mRNA-Luc-loaded LNPs at an mRNA dose of 0.2 mg/kg. Images were acquired at 4h post-treatment. Below the images, a table with the formulation details of LNPs and pie charts that show a profile of the tissue-specificity of mLuc expression in the lungs, heart, liver, intestine, spleen and kidneys and the quantification of the majority percent expression detected in each LNP.

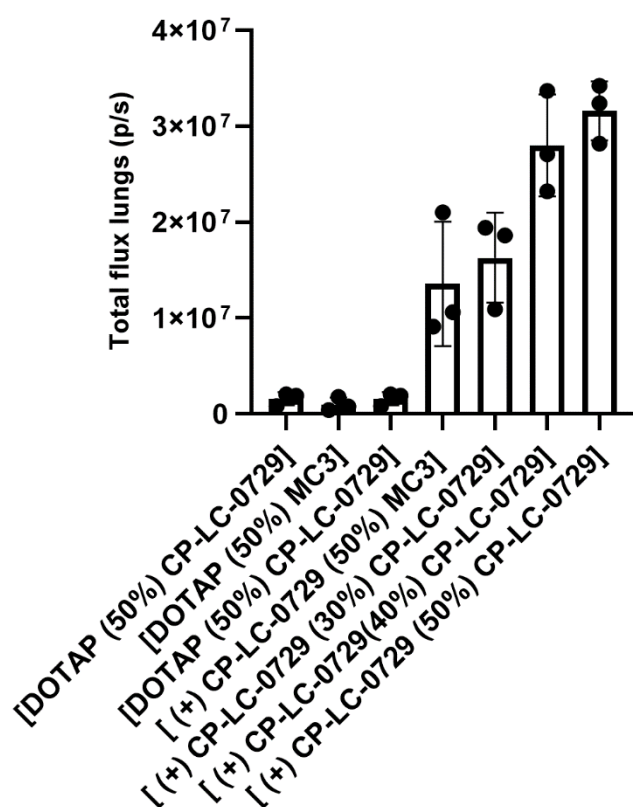


Figure S6. Quantification of the total luminescence flux produced by the LNPs containing cationic lipids in the lungs (n= 3 biologically independent samples). Mice were i.v injected with mRNA-Luc-loaded LNPs at an mRNA dose of 0.2 mg/kg. Images and their quantification were taken at 4h post-treatment. Data are presented as mean values $\pm$ SD. One-way ANOVA with Tukey's correction was used.

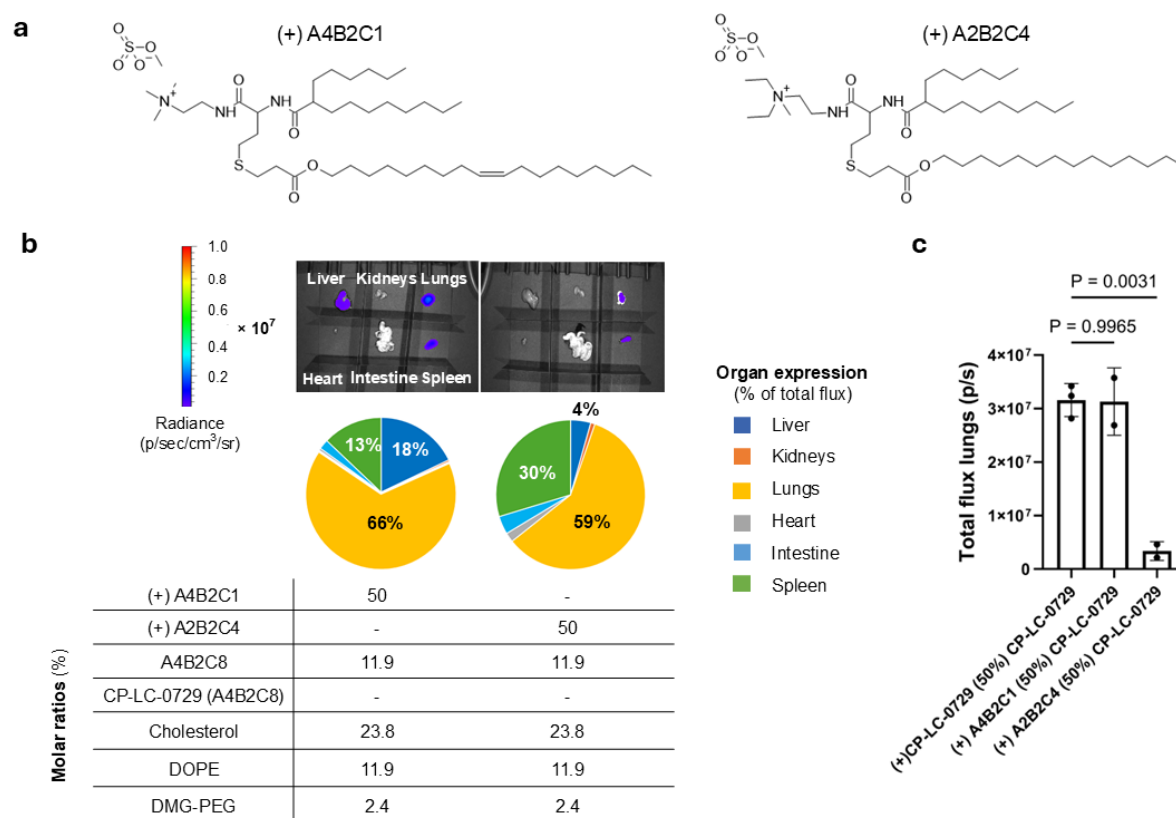


Figure S7. a. Chemical structures of cationic lipids (+) A4B2C1 and (+) A2B2C4. **b.** *Ex vivo* luminescence of major organs from mice and their quantification (n= 2 biologically independent samples). Mice were i.v. injected with mRNA-Luc-loaded LNPs at an mRNA dose of 0.2 mg/kg. Images were acquired at 4h post-treatment. Below the images, a table with the formulation details of LNPs and pie charts that show a profile of the tissue-specificity of mLuc expression in the lungs, heart, liver, intestine, spleen and kidneys and the quantification of the most significant expression targets detected in each LNP. **c.** Quantification of the total luminescence flux produced by the LNPs containing cationic lipids in the lungs (n= 2 biologically independent samples). Mice were i.v injected with mRNA-Luc-loaded LNPs at an mRNA dose of 0.2 mg/kg. Images and their quantification were taken at 4h post-treatment. Data are presented as mean values ± SD. One-way ANOVA with Tukey's correction was used.

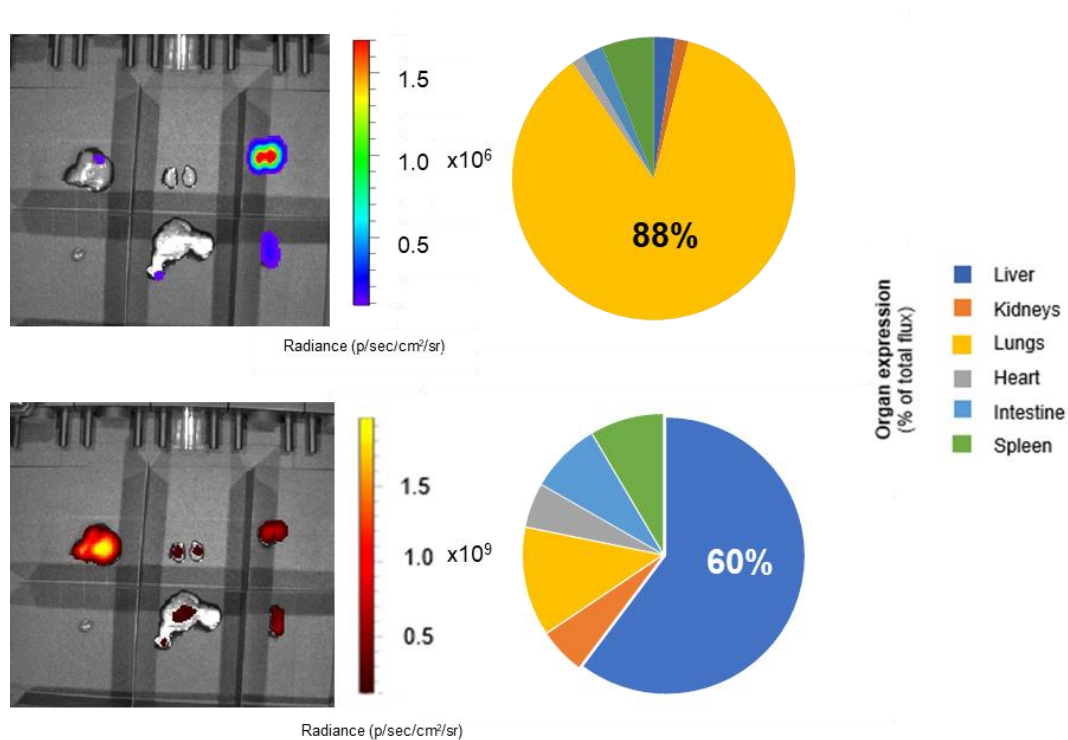


Figure S8. Images of the luminescence (above) and fluorescence (below) produced by [(+) CP-LC-0729 (50%) CP-LC-0729]-DiR LNP in the liver, kidneys, lungs, heart, intestine and spleen (n= 3 biologically independent samples). Mice were i.v. injected with mLuc-loaded LNPs at an mRNA dose of 0.2 mg/kg. Images were taken at 4h post-treatment. Beside the images, pie charts profile the tissue-specificity of mLuc (above) and fluorescence (below) expression in the lungs, heart, liver, intestine, spleen and kidneys and quantify the majority percent expression detected.

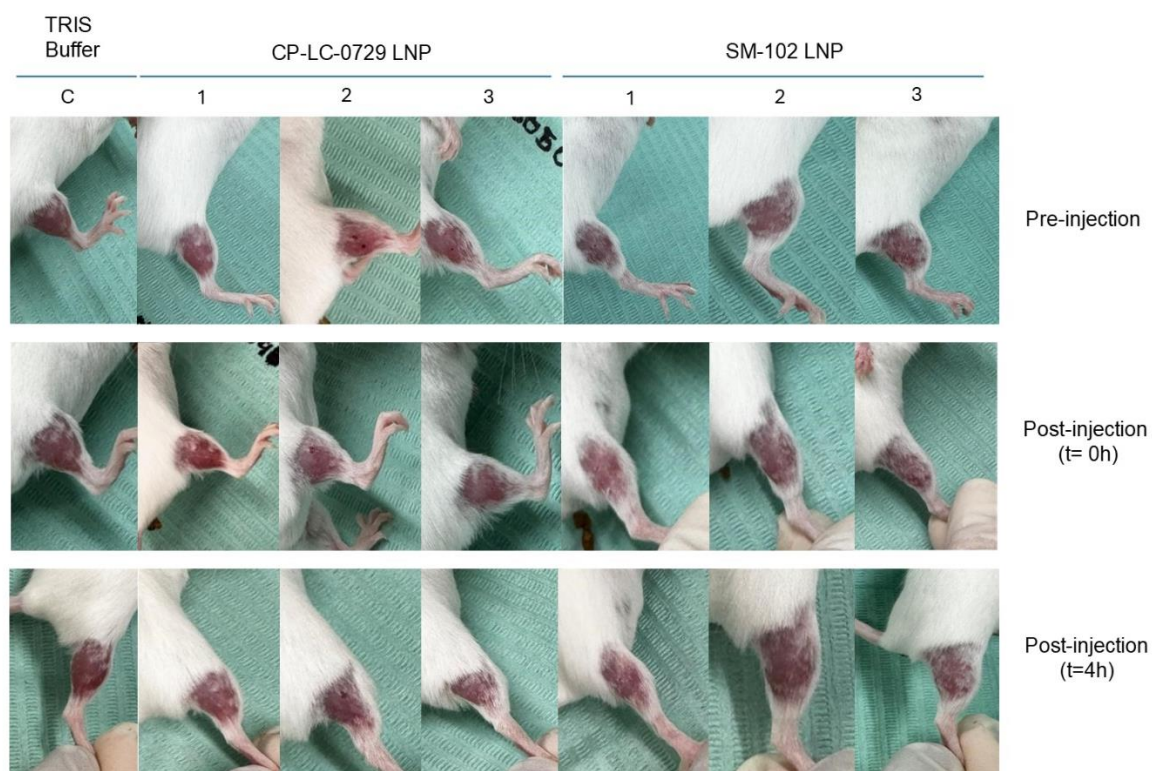


Figure S9. Images of the intramuscular toxicity experiment comparing the effects before, immediately after and after 4h of inoculation of TRIS buffer, CP-LC-0729 LNP and SM-102 LNP. Mice were i.m. injected with mRNA-Luc-loaded LNPs at an mRNA dose of 0.3 mg/kg.

Table S1. A compilation of the detected m/z values in the mass spectra of STAAR Lipids during two optimization phases of screening:

		Theoretical [M+H] ⁺	Experimental [M+H] ⁺
Phase 1 optimization	A1B1C1	808.69	808.84
	A1B1C2	670.55	670.73
	A1B1C3	726.62	726.81
	A1B1C4	754.64	754.83
	A1B1C5	726.61	726.84
	A1B1C6	670.55	670.72
	A1B1C7	670.55	670.76
	A1B2C1	780.66	780.80
	A1B2C2	642.52	642.68
	A1B2C3	698.60	698.76
	A1B2C4	726.61	726.82

	A1B2C5	698.58	698.82
	A1B2C6	642.52	642.67
	A1B2C7	642.52	642.69
	A1B3C1	806.68	806.82
	A1B3C2	668.53	668.73
	A1B3C3	724.60	724.80
	A1B3C4	752.63	752.87
	A1B3C5	724.60	724.85
	A1B3C6	668.53	668.73
	A1B3C7	668.53	668.73
Phase 2 optimization	A2B2C1	794.68	794.87
	A2B2C2	656.53	656.69
	A2B2C3	712.60	712.79
	A2B2C4	740.63	740.82
	A2B2C5	712.60	712.79
	A2B2C6	656.53	656.70
	A2B2C7	656.53	656.70
	A3B2C1	780.66	780.82
	A3B2C2	642.52	642.64
	A3B2C3	698.59	698.80
	A3B2C4	726.61	726.82
	A3B2C5	698.58	698.78
	A3B2C6	642.03	642.67
	A3B2C7	642.52	642.67
	A4B2C1	766.25	766.81
	A4B2C2	628.50	628.66
	A4B2C3	684.57	684.75
	A4B2C4	712.60	712.74
	A4B2C5	684.57	684.74
	A4B2C6	628.50	628.65
	A4B2C7	628.50	628.65
	A5B2C1	821.69	821.84
	A5B2C2	683.55	683.70
	A5B2C3	739.61	739.81
	A5B2C4	767.64	767.81
	A5B2C5	739.61	739.79
	A5B2C6	683.55	683.71
	A5B2C7	683.55	683.70
	A6B2C1	777.62	777.77
	A6B2C2	665.50	665.71
	A6B2C3	721.57	721.77
	A6B2C4	749.59	749.86
	A6B2C5	721.56	721.79
	A6B2C6	665.50	665.69
	A6B2C7	665.50	665.71
	A7B2C1	822.31	822.86

	A7B2C2	684.53	684.72
	A7B2C3	740.60	740.79
	A7B2C4	768.62	768.80
	A7B2C5	740.59	740.80
	A7B2C6	684.53	684.70
	A7B2C7	684.53	684.68
	A8B2C1	792.66	792.78
	A8B2C2	654.52	654.68
	A8B2C3	710.59	710.77
	A8B2C4	738.61	738.83
	A8B2C5	710.58	710.77
	A8B2C6	654.52	654.69
	A8B2C7	654.52	654.69
	A9B2C1	806.68	806.84
	A9B2C2	668.53	668.73
	A9B2C3	724.60	724.78
	A9B2C4	752.63	752.85
	A9B2C5	724.60	724.83
	A9B2C6	668.53	668.74
	A9B2C7	668.53	668.73
	A10B2C1	840.69	840.86
	A10B2C2	702.54	702.74
	A10B2C3	758.60	758.79
	A10B2C4	786.64	786.83
	A10B2C5	758.61	758.80
	A10B2C6	702.54	702.73
	A10B2C7	702.54	702.75
	A11B2C1	806.68	806.85
	A11B2C2	668.54	668.73
	A11B2C3	724.60	724.81
	A11B2C4	752.63	752.83
	A11B2C5	724.60	724.82
	A11B2C6	668.54	668.71
	A11B2C7	668.54	668.72

Table S2. Characterization of formulated LNPs in screening phases 1 and 2.

		Particle size (nm)	P.D.I.	Zeta potential (mV)	%EE
Phase 1 screening	A1B1C1	221.2	0.26	9.1	98.4
	A1B1C2	184.5	0.18	8.9	99.7
	A1B1C3	185.2	0.28	6.9	96.5
	A1B1C4	232.3	0.32	8.1	98.6
	A1B1C5	247.8	0.24	8.3	98.3
	A1B1C6	183.2	0.19	8.0	99.7
	A1B1C7	177.3	0.18	9.4	99.8
	A1B2C1	262.1	0.24	4.8	97.4
	A1B2C2	189.2	0.14	-6.1	92.3
	A1B2C3	142.9	0.11	-6.8	89.5
	A1B2C4	271.4	0.25	4.1	98.6
	A1B2C5	208.4	0.16	0.6	98.9
	A1B2C6	215.1	0.19	-1.4	98.7
	A1B2C7	209.9	0.16	0.7	97.3
	A1B3C1	297.3	0.36	8.5	97.4
	A1B3C2	190.7	0.21	11.7	99.6
	A1B3C3	175.6	0.21	5.1	98.9
	A1B3C4	264.2	0.32	11.0	98.7
	A1B3C5	240.7	0.23	13.7	98.7
	A1B3C6	206.4	0.18	13.4	99.4
	A1B3C7	204.0	0.19	12.5	99.7
Phase 2 screening	A2B2C1	143.8	0.13	-5.9	60.6
	A2B2C2	148.5	0.23	-9.1	35.5
	A2B2C3	145.1	0.10	-9.4	50.0
	A2B2C4	156.2	0.17	-7.0	65.2
	A2B2C5	146.7	0.15	-9.6	57.4
	A2B2C6	161.6	0.21	-5.9	64.0
	A2B2C7	136.9	0.16	-6.1	56.2
	A3B2C1	193	0.12	-12.3	76.3
	A3B2C2	207.1	0.14	-11.1	49.7
	A3B2C3	165	0.25	-13.4	50.0
	A3B2C4	191.9	0.15	-11.5	75.1
	A3B2C5	186.3	0.14	-7.5	80.1
	A3B2C6	143.8	0.17	-16.3	57.8
	A3B2C7	203.8	0.16	-10.2	65.1
	A4B2C1	225.8	0.17	-1.4	73.2
	A4B2C2	171.9	0.13	-3.0	67.4
	A4B2C3	109.1	0.15	-8.5	95.9
	A4B2C4	201.9	0.19	-1.7	65.2
	A4B2C5	204.3	0.17	-0.5	44.4
	A4B2C6	197.0	0.11	-0.4	80.1
	A4B2C7	181.7	0.12	-2.2	68.1

	A5B2C1	231.7	0.31	4.9	98.7
	A5B2C2	206.7	0.35	-6.5	85.5
	A5B2C3	148.2	0.14	-7.2	81.3
	A5B2C4	186.8	0.15	-1.6	92.4
	A5B2C5	217.1	0.22	-2.7	91.2
	A5B2C6	135.2	0.11	4.8	87.0
	A5B2C7	158.7	0.18	3.4	86.7
	A6B2C1	211.3	0.10	-7.4	47.1
	A6B2C2	138.8	0.14	-8.5	41.9
	A6B2C3	132.0	0.14	-10.9	44.3
	A6B2C4	187.3	0.13	-12.6	58.1
	A6B2C5	161.7	0.18	-17.2	44.3
	A6B2C6	176.7	0.15	-10.3	36.6
	A6B2C7	147.0	0.15	-8.3	39.0
	A7B2C1	169.3	0.18	-12.2	41.0
	A7B2C2	164.8	0.11	-24.6	42.5
	A7B2C3	190.3	0.27	-10.6	51.0
	A7B2C4	198.1	0.16	-23.7	58.8
	A7B2C5	200.6	0.22	-22.3	34.3
	A7B2C6	204.1	0.17	-9.7	44.9
	A7B2C7	228.8	0.24	-23.1	53.7
	A8B2C1	216.2	0.23	1.7	84.5
	A8B2C2	179.0	0.27	-20.9	83.3
	A8B2C3	135.9	0.12	-11.1	69.8
	A8B2C4	174.1	0.11	-0.9	87.2
	A8B2C5	221.3	0.19	4.5	64.6
	A8B2C6	196.7	0.13	-7.0	92.8
	A8B2C7	156.7	0.15	-2.4	72.8
	A9B2C1	146.9	0.13	1.0	93.4
	A9B2C2	175.0	0.16	1.1	53.6
	A9B2C3	151.2	0.12	-1.0	92.3
	A9B2C4	192.7	0.16	10.0	92.7
	A9B2C5	177.3	0.16	5.9	94.6
	A9B2C6	168.7	0.20	7.0	80.0
	A9B2C7	142.0	0.21	3.7	63.7
	A10B2C1	184.8	0.16	2.5	97.5
	A10B2C2	153.3	0.20	-8.2	87.1
	A10B2C3	158.4	0.12	-9.3	71.7
	A10B2C4	194.1	0.16	-1.6	96.0
	A10B2C5	227.4	0.29	-2.9	90.2
	A10B2C6	185.7	0.20	-0.8	93.6
	A10B2C7	171.7	0.14	0.8	89.6
	A11B2C1	142.7	0.12	0.3	90.8
	A11B2C2	156.1	0.11	-11.2	43.9
	A11B2C3	229.1	0.20	-1.4	72.0

	A11B2C4	163.6	0.13	1.2	87.6
	A11B2C5	193.9	0.21	-2.1	87.5
	A11B2C6	223.3	0.27	-11.1	74.8
	A11B2C7	251.3	0.32	-14.0	47.3

LNPs were formulated by pipetting mixing. Aqueous and organic solutions were mixed by pipetting the aqueous solution into the organic solution by vigorous pipetting for about 10 seconds. The resulting lipid particles encapsulating mRNAs were immediately neutralized with a Tris buffered solution with 15% sucrose. Lipid particles were characterized by measuring their size distribution, polydispersity and Z-potential by dynamic light scattering (DLS). mRNA encapsulation was assessed by Quant-IT® Ribogreen following the manufacturer's instructions.

Table S3. LNP formulation controls used.

	Components	Molar Ratio	N/P ratio (mol/mol)
MC3 Benchmark	MC3/DSPC/Chol/DMG-PEG	50:10:38.5:1.5	6
MC3 (DOPE)	MC3/DOPE/Chol/DMG-PEG	50:10:38.5:1.5	6
SM-102	SM-102/DOPE/Chol/DMG-PEG	50:10:38.5:1.5	6

Table S4. Characterization of formulated LNPs by microfluidics after screening phases.

	Particle size (nm)	P.D.I.	Zeta potential (mV)	Apparent pKa	%EE
A4B4C3	133.3	0.18	-5.9	6.72	90.2
A4B4C8	127.4	0.09	-6.6	6.96	97.4
A4B4C9	104.8	0.15	-5.7	7.01	96.2
A4B2C3	109.1	0.15	-8.4	6.82	96.0
A4B2C8 (CP-LC-0729)	91.5	0.15	-7.7	6.78	95.1
A4B2C9	86.1	0.12	-9.1	6.89	94.1
A4B5C3	107.8	0.20	-5.3	6.76	97.5
A4B5C8	98.5	0.26	-10.4	6.63	96.5
A4B5C9	90.3	0.14	-7.8	6.61	94.4

MC3 (DOPE)	124.6	0.22	-11.0	6.40	92.0
MC3 (DSPC)	86.5	0.28	-13.0	6.15	96.3
SM-102	92.9	0.07	-4.9	6.95	92.5
CP-LC-0729-01	72.7	0.12	-10.3	6.12	93.8
CP-LC-0729-02	90.6	0.09	-8.2	6.81	96.1
CP-LC-0729-03	76.3	0.11	-5.9	7.06	97.1
CP-LC-0729-04	87.7	0.15	-9.5	7.18	98.0
CP-LC-0729-05	80.6	0.25	-18.0	5.48	92.9

LNPs were formulated by microfluidic mixing. The resulting lipid particles encapsulating mRNAs were dialyzed overnight against a pH 8 Tris buffer solution containing 15% sucrose. Lipid particles were characterized by measuring their size distribution, polydispersity and Z-potential by dynamic light scattering (DLS). mRNA encapsulation was assessed by Quant-IT® Ribogreen following the manufacturer's instructions. Apparent pKa was calculated using a 6-(p-toluidinyl)naphthalene-2-sulfonic acid (TNS) assay.

Table S5. LNP formulation used in lung targeting strategy by microfluidics.

	Components	Molar Ratio	N/P ratio (w/w)
[DOTAP (50%) CP-LC-0729]	DOTAP/ CP-LC-0729 /DOPE/Chol/DMG-PEG	50/11.9/11.9/23.8/2.4	10
[(+) CP-LC-0729 (50%) MC3]	(+) CP-LC-0729 /MC3/DOPE/Chol/DMG-PEG	50/11.9/11.9/23.8/2.4	10
[(+) CP-LC-0729 (30%) CP-LC-0729]	(+) CP-LC-0729 / CP-LC-0729 /DOPE/Chol/DMG-PEG	30/16.7/16.7/33.3/3.4	10
[(+) CP-LC-0729 (40%) CP-LC-0729]	(+) CP-LC-0729 / CP-LC-0729 /DOPE/Chol/DMG-PEG	40/14.3/14.3/28.6/2.9	10
[(+) CP-LC-0729 (50%) CP-LC-0729]	(+) CP-LC-0729 / CP-LC-0729 /DOPE/Chol/DMG-PEG	50/11.9/11.9/23.8/2.4	10
[(+) CP-LC-0729 (50%) CP-LC-0729]-DiR	(+) CP-LC-0729 / CP-LC-0729 /DOPE/Chol/DMG-PEG/DiR	49.9/11.8/11.8/23.7/2.4/0.4	10
[(+) A2B2C4 (50%) CP-LC-0729]	(+) A2B2C4 / CP-LC-0729 /DOPE/Chol/DMG-PEG	50/11.9/11.9/23.8/2.4	10
[(+) A4B2C1 (50%) CP-LC-0729]	(+) A4B2C1 / CP-LC-0729 /DOPE/Chol/DMG-PEG	50/11.9/11.9/23.8/2.4	10

Table S6. Characterization of formulated LNPs by microfluidics for lung targeting strategy.

	Particle size (nm)	P.D.I.	Zeta potential (mV)	Apparent pKa	%EE
[DOTAP (50%) MC3] LNP	58.68	0.1975	14.71	7.29	98.84
[DOTAP (50%) CP-LC-0729]	66.6	0.25	12.15	6.16	98.9
[(+) CP-LC-0729 (50%) MC3]	78.6	0.37	7.13	8.16	99.2
[(+) CP-LC-0729 (30%) CP-LC-0729]	65.9	0.26	6.48	7.83	98.6
[(+) CP-LC-0729 (40%) CP-LC-0729]	81.9	0.35	9.366	7.93	99.1
[(+) CP-LC-0729 (50%) CP-LC-0729]	71.7	0.30	11.54	7.37	99.2
[(+) CP-LC-0729 (50%) CP-LC-0729]d	60.9	0.23	13.65	7.76	99.6
[(+) A2B2C4 (50%) CP-LC-0729]	57.83	0.20	18.82	n/a	98.4
[(+) A4B2C1 (50%) CP-LC-0729]	91.77	0.25	11.14	8.64	99.6

LNPs were formulated by microfluidic mixing. The resulting lipid particles encapsulating mRNAs were dialyzed overnight against a pH 8 Tris buffer solution containing 15% sucrose. Lipid particles were characterized by measuring their size distribution, polydispersity and Z-potential by dynamic light scattering (DLS). mRNA encapsulation was assessed by Quant-IT® Ribogreen following the manufacturer's instructions. Apparent pKa was calculated using a 6-(p-toluidinyl) naphthalene-2-sulfonic acid (TNS) assay.

Table S7. Characterization and in vivo performance of formulated LNPs of lipid CP-LC-0729 in different lipid molar ratios by microfluidics.

Lipid Molar Ratios (CP-LC-0729/DOPE/Chol/DMG-PEG)	Size (nm)	PDI	Zeta potential (mV)	EE (%)	Total Flux (p/s)
40/16/40/4	67.4	0.2679	-3.135	98.0	7.15E+07
35.7/28.6/28.6/7.1	57.5	0.1544	-7.321	91.0	2.21E+07
27.6/33.1/38.7/0.6	122.6	0.1760	-9.176	96.8	8.34E+08

26/41.7/31.3/1	108.8	0.2087	-6.754	96.3	2.45E+08
43.8/12.5/37.5/6.3	62.34	0.2737	-6.922	93.4	3.11E+07
37.8/21.6/37.8/2.7	88.73	0.3088	-12.28	95.8	1.45E+08
40.7/34.9/23.3/1.2	103.4	0.2180	-8.246	95.7	8.50E+08
34.8/39.8/24.9/0.5	151.2	0.0571	-6.022	94.7	1.03E+09
42.9/28.6/23.8/4.8	75.89	0.3078	-13.32	81.8	5.27E+07
41.9/37.2/18.6/2.3	85.5	0.2122	-8.296	89.4	1.63E+08
47.8/26.1/21.7/4.3	74.55	0.1584	-12.5	70.4	1.16E+08
46.8/34/17/2.1	97.23	0.2288	-8.555	86.7	1.95E+08
50/10/38.5/1.5	98.47	0.2114	-9.349	91.3	7.05E+08

LNPs were formulated by microfluidic mixing at N/P ratios of 6 (mol/mol). The resulting lipid particles encapsulating mRNAs were dialyzed overnight against a pH 8 Tris buffer solution containing 15% sucrose. Lipid particles were characterized by measuring their size distribution, polydispersity and Z-potential by dynamic light scattering (DLS). mRNA encapsulation was assessed by Quant-IT® Ribogreen following the manufacturer's instructions. Apparent pKa was calculated using a 6-(p-toluidinyl) naphthalene-2-sulfonic acid (TNS) assay. Mice were i.m injected with mRNA-Luc-loaded LNPs at an mRNA dose of 0.05 mg/kg. Luminescence imaging acquisition was performed at 4 h post-treatment and total flux was quantified. In vivo mLuc expression (n= 3 biologically independent samples). Data are presented as mean values.

References

1. Gabiras Bonnie J. & Marburg Stephen. Preparation of Carboxythiolactones and Their Active Derivatives. *Synthesis* **2**, 270-274 (1999).
2. Broset Blasco, E. & Martínez Oliván, J. E. Artificial polynucleotides for expressing proteins. EP4349990A1 (2022).